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### BSAI Final Year Project Portfolio — Volume 2

#### 10 Evaluated AI/ML FYP Ideas: Deepfake & Media Forensics + Non-Banking FinTech, with Visual/Interactive Demos

Prepared as: Senior AI Researcher / Startup Advisor / FYP Evaluator Assessment Target: BSAI Undergraduate Team (2-4 members), 8-10 Month Timeline, Software-Only, Visually Demoable

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#### Grounding Note: What Is Actually True About Deepfake Detection as of 2026

Before the ideas, here is the real state of the field your PulseGuardAI inspiration sits inside, since you specifically asked what work has been done through 2026 and what its limitations are. This is not a generic “deepfakes are a problem” framing; these are the specific, current, citable failure modes your FYP can target.

**rPPG-based physiological detection (your inspiration’s core idea) is a real, active research direction, not a novel idea you’d be inventing from scratch.** A 2026 IEEE review confirms rPPG is considered a powerful deepfake-detection tool specifically because deepfake videos often fail to replicate the physiological signals inherent in real human subjects. Multiple 2025-2026 papers (AVENUE/TCN+rPPG, STFT phase-feature rPPG, Forensic Lens graph-based rPPG) confirm this is an active, current sub-field, which is good news: there’s a real literature to build on, cite, and differentiate from, rather than an empty space.

**But the documented limitations are specific and exploitable for your novelty:** - A 2025 systematic review found that only 46.3% of selected publications believed their deepfake-detection techniques could generalize across different deepfake types, with overfitting to the specific training dataset identified as the key limitation reducing effectiveness on new or unseen fakes. - The rPPG signal itself is fragile: rPPG-based detection can be inaccurate when the signal is corrupted by facial-deformation noise and illumination variation, and most existing models use sequential architectures that fail to properly process long temporal sequences — this is exactly the gap your inspiration’s “Temporal Transformer” idea targets, confirming it’s a real, current research direction, not a toy idea. - Existing physiological/rPPG methods often reveal scalability shortcomings, sensitivity to dataset bias, and demand computational resources that are prohibitive for real-world deployment — meaning a genuinely lightweight rPPG pipeline is itself a contribution, not just an implementation detail. - The single most important 2025-2026 finding for your evaluator pitch: models trained on controlled academic datasets (FaceForensics++, ASVspoof, Celeb-DF) generalize poorly to real in-the-wild deepfakes, with even strong academic models losing up to half their discriminative

power on transfer, and the best commercial detectors still only reach roughly 78% accuracy, well below the approximately 90% accuracy of trained human forensic analysts. This is the generalization gap your FYP should explicitly measure and try to narrow, not just another accuracy leaderboard entry. - Diffusion/text-to-video generation (Sora, Runway-Gen3-class models) breaks an entire category of prior forensic methods: frame-pair differential anomaly detectors cannot operate on completely AI-generated single frames, and newer text-to-video and image-to-video models can produce ultra-realistic fake videos from a single prompt or starting frame, which older face-swap-era detectors were never designed to catch. - Audio deepfakes have an even fresher, more exploitable gap: none of the standard audio-deepfake benchmarks (ASVspoof 2021, Deepfake-in-the-Wild, Deepfake-Eval-2024) contain audio from the newest 2024-2025 TTS architectures, and emotional, non-neutral speech (laughing, sighing, whispering) is almost entirely unstudied as a detection challenge despite being increasingly common in modern TTS output. - The most encouraging trend for your GPU-constrained team: the May 2026 literature explicitly favors small, targeted forensic cues over giant backbones — a tiny 292-parameter fusion module combining wavelet-denoised low-frequency features with phase/texture cues raised AUC from 74.8% to 78.6% on FaceForensics++, outperforming larger established dual-stream detectors like F3Net, SRM, and SPSL across eight benchmarks, and the same month's literature found frozen vision foundation-model backbones remain discriminative for full-face synthesis but reveal real limits on localized face edits. The lesson for your team: a small, well-designed, well-evaluated detector that's honest about its generalization boundary will impress evaluators far more than a large model with an inflated single-dataset accuracy number.

These findings are woven directly into the novelty-analysis section of each relevant idea below, not just summarized here, so you can cite them specifically when your evaluator asks “what’s actually new about this.”

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## Evaluation Methodology

Every idea below was filtered against your explicit constraints: deepfake/media-forensics or non-banking finance only; software-only (no hardware/IoT/sensors); 8-10 months for a 2-4 person undergraduate team; trainable on a single consumer GPU or free Colab/Kaggle; real public datasets; a strong visual/interactive demo component (since you were explicit that dry, dashboard-less academic projects don't satisfy your style requirement); and no generic chatbot, recommendation system, or basic CRUD+ML wrapper.

A few commonly-pitched ideas were deliberately excluded, and the reasoning matters for your decision-making. A generic “deepfake image classifier” trained end-to-end on FaceForensics++ alone was excluded because it has no defensible novelty left in 2026; thousands of such models exist and the benchmark is close to saturated for the easy cases. A “crypto price prediction LSTM” was excluded because crypto price series are close to a random walk at short horizons and naive prediction projects rarely produce honest, defensible results; the related ideas below instead focus on market *behavior* and *risk visualization*, which is both more honest and more visually interesting. A “voice cloning detector using only spectrograms + CNN” was excluded as

the single most repeated audio-deepfake FYP topic; the audio idea below specifically targets the 2024-2025 TTS generalization gap instead, which is a meaningfully different and fresher angle.

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## IDEA 1: Multi-Region Physiological Synchrony Deepfake Detector (Direct Evolution of Your Inspiration)

### 1. Problem Statement

Deepfake videos are increasingly used for disinformation, financial fraud (CEO voice/video impersonation scams), and harassment, and existing pixel-artifact-based detectors fail against new generation methods almost as fast as they're trained, because they learn dataset-specific compression and rendering artifacts rather than anything fundamentally tied to "is this a real human." The people affected are journalists, platform trust-and-safety teams, forensic investigators, and ordinary people targeted by impersonation scams. This matters because the documented generalization gap between controlled-benchmark performance and in-the-wild performance is currently severe, and any method that narrows it, even partially, is a genuine contribution.

### 2. Proposed Solution

This is your PulseGuardAI inspiration, kept largely intact but scoped and strengthened with the specific 2026 research context now built in. The core mechanism: extract rPPG (remote photoplethysmography) signals from multiple facial regions (forehead, left cheek, right cheek, and optionally the chin/jaw region for a fourth comparison point), and check both signal plausibility (does this look like a real cardiac waveform) and, critically, the multi-region *synchrony* (do all regions beat in phase, as they would in a real person, where blood flow is driven by one shared cardiac cycle).

This is visually and interactively excellent for a demo: you upload a video and watch four live waveform traces animate in sync (or visibly drift apart) on screen, with a synchrony-mismatch score displayed as a number and a color-coded alert. This is exactly the kind of "you can see the AI's reasoning happen" system your style requirement asks for, and it is fundamentally more interesting in a viva than a single "fake: 87%" probability number.

### 3. Technical Architecture

Data sources: FaceForensics++ (real + four manipulation types, multiple compression levels), Celeb-DF v2 and the newer, more diverse Celeb-DF++ (22 manipulation methods across face-swap/reenactment/talking-face categories) for cross-method generalization testing, and critically, DeepFake-Eval-2024 (a 2025-released, freely downloadable, in-the-wild benchmark specifically built to expose the controlled-dataset generalization gap) as your held-out, "did we actually solve anything" final test set.

ML/DL models: a face-tracking and ROI (region of interest) extraction stage (MediaPipe or a lightweight face-mesh model, both CPU-feasible and well-documented); a CHROM or POS-

based classical rPPG signal-extraction algorithm per region (no training needed, this is signal processing, a legitimate and honest part of the pipeline, not everything needs to be a neural net); a lightweight temporal model (a small Temporal Convolutional Network or a compact Transformer encoder, deliberately kept small given the 2026 literature's finding that tiny, targeted models can outperform large backbones) that takes the four region-wise rPPG waveforms plus their pairwise phase-difference and cross-correlation features, and outputs both a real/fake classification and the synchrony-mismatch score as an interpretable intermediate output, not just a hidden activation.

Training pipeline: extract and cache rPPG waveforms for the full training set first (this is the slow, CPU-bound preprocessing step, budget real time for it); train the lightweight temporal classifier on the extracted waveform-and-synchrony features; use FaceForensics++ as primary training data, with explicit cross-dataset evaluation (never train and test on the same dataset only) against Celeb-DF++ and DeepFake-Eval-2024 to honestly measure the generalization gap your inspiration's note already anticipated.

Evaluation pipeline: standard AUC/accuracy on FF++ held-out test split (the easy, expected number); cross-dataset AUC on Celeb-DF++ and DeepFake-Eval-2024 (the number that actually matters and that most undergraduate FYPs never report, making its inclusion itself a credibility signal to evaluators); an explicit ablation comparing single-region rPPG detection against your multi-region synchrony approach, which is the paper's central empirical claim and must be reported even if the improvement is modest, since an honest modest improvement, well-measured, beats an inflated unverified claim.

Deployment architecture: a web app (React frontend, FastAPI backend) where a user uploads a video and watches, in near-real-time, the four ROI boxes tracked on the face, the four waveforms animating below, a live synchrony-mismatch gauge, and a final verdict with the specific physiological reasoning stated in plain language (e.g., "forehead-cheek phase lag of 60ms exceeds the 20ms threshold observed in real video training data"), directly fulfilling your explainability requirement.

#### 4. Novelty Analysis

Existing solutions: published rPPG-deepfake-detection work (AVENUE/TCN, STFT-phase, Forensic Lens) largely treats rPPG as a single global facial signal or compares limited region pairs, and per the 2025 systematic review, the field-wide finding is that generalization across unseen deepfake types remains an unresolved limitation, with overfitting to the training dataset identified as the central cause. Your innovation: treating multi-region phase-synchrony (not just waveform plausibility) as the primary signal, explicitly because synchrony is a physiologically harder property for a generative model to accidentally reproduce than a single plausible-looking waveform, combined with rigorous, explicit cross-dataset generalization reporting against the 2025-released DeepFake-Eval-2024 in-the-wild benchmark, which most published rPPG papers (predating its release) could not have tested against. This combination, a specific physiological signal choice plus honest generalization measurement against the newest in-the-wild benchmark, is a genuinely defensible, current contribution.

## 5. Dataset Availability

Public datasets: FaceForensics++ (apply for access via the official project page, academic email required, approval is typically fast, standard procedure well-documented in many FYP and thesis projects), Celeb-DF v2/++ (similarly requires an academic-use request form, standard and fast), DeepFake-Eval-2024 (fully public, no application needed, directly downloadable from its GitHub repository). Data collection strategy: no manual collection needed, all three datasets are sufficient; the team's own webcam-recorded "real" videos can supplement as an easy, bias-check sanity dataset (different lighting/skin tones/cameras than the academic datasets) at near-zero cost. Estimated dataset size needed: a few thousand video clips total across training and cross-dataset evaluation, well within what FF++/Celeb-DF++ provide; the team does not need the full multi-hundred-thousand-clip scale of these datasets, a curated subset is entirely sufficient and faster to iterate on.

## 6. Development Complexity

Difficulty: Medium-Hard (the rPPG-extraction and synchrony-feature engineering is the genuinely hard, research-adjacent part; the classifier itself is small and easy to train). Estimated completion time: 8-9 months. GPU requirements: low; the temporal classifier is small (per the 2026 literature trend toward lightweight models) and trains comfortably on a single consumer GPU (6-8GB) or free Colab/Kaggle T4 in hours, not days; the rPPG extraction itself is CPU-bound signal processing, not GPU-bound. Team size suitability: ideal for 3 students (one on face-tracking/ROI/signal-extraction pipeline, one on the temporal classifier and cross-dataset evaluation, one on the live interactive dashboard).

## 7. Funding Potential

Why investors/incubators/grants may care: deepfake detection for trust-and-safety, fraud prevention (CEO-voice/video-fraud detection is now a recognized corporate security category), and content-platform moderation is an actively growing, internationally fundable deep-tech category; in Pakistan specifically, this fits Ignite's stated focus on deep-tech innovation grants and is a strong fit for the HEC Technology Development Fund's university-research-commercialization framing given the genuine algorithmic novelty involved. Potential market size: global; every video-conferencing platform, content moderation team, and corporate-security vendor is a plausible eventual customer or partner. Commercialization opportunities: an API-as-a-service for video-authenticity scoring (sold to platforms or security vendors), or a browser-extension consumer-facing demo (e.g., "verify this video clip") as a lighter-weight go-to-market that's far more demoable in a pitch than an enterprise sales motion.

## 8. Research Potential

Strong. A paper on "multi-region physiological synchrony as a generalizable deepfake forensic cue: a cross-dataset evaluation" is directly publishable-track, fits an applied-AI or multimedia-forensics venue (workshops at ICASSP, WIFS - the IEEE Workshop on Information Forensics and Security, or a CVPR/WACV workshop on media forensics, all realistic, undergraduate-reachable targets if the cross-dataset evaluation is rigorous). Research questions: does multi-region synchrony degrade less under video compression than single-region waveform plausibility (a clean, specific, testable hypothesis tied directly to real-world deployment

conditions like WhatsApp/social-media compression); how does synchrony-based detection perform specifically on talking-face and reenactment forgeries (Celeb-DF++'s TF category) versus classic face-swap, since the 2025 Celeb-DF++ paper found robust cross-dataset results are largely limited to face-swap forgeries, with face-reenactment and talking-face forgeries yielding substantially degraded detection rates, making this a genuinely open and citable question your project could help answer.

## 9. Risks and Challenges

Technical risks: rPPG extraction is sensitive to lighting, head motion, and video compression, exactly the real-world conditions your evaluation should test, not avoid; the team must budget real time for robustness testing rather than only reporting results on clean, well-lit academic footage. Data risks: minimal, dataset access is the only friction point and is a known, fast, standard process. Deployment risks: a deepfake detector that's wrong, in either direction, has real consequences (falsely clearing a real deepfake fraud video, or falsely flagging a real video); the tool should be explicitly framed and demoed as a forensic-assistance score for a human reviewer, not an autonomous verdict, and this framing should be visible in the dashboard UI itself, not just the written report.

## 10. Final Verdict

Innovation: 8/10. Practicality: 7/10. Funding Potential: 7/10. Research Potential: 9/10. Ease of Completion: 6/10.

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# IDEA 2: Diffusion-Era Single-Frame Forensic Localizer (Targeting the Sora/Runway Generalization Gap)

## 1. Problem Statement

The deepfake detection methods that dominate current academic literature (pixel-artifact CNNs, frame-pair differential methods, even rPPG-based approaches) were largely designed for face-swap and reenactment forgeries, but a structurally different and newer threat has emerged: fully AI-generated video from diffusion-based text-to-video and image-to-video models (Sora-class systems), where there is no "original real frame" to compare against at all, since the entire frame is synthesized from a text prompt or single starting image. Existing frame-pair and rPPG methods, including Idea 1 above, are not designed for this case and a 2025 paper explicitly flags this as an open gap. This affects the same population as Idea 1, content platforms, forensic investigators, and journalists, but for a meaningfully different and newer threat category that current tooling has not caught up to.

## 2. Proposed Solution

A spatial-forensic localizer that operates on single frames (not requiring a real-vs-fake frame pair or a temporal pulse signal) and is specifically trained and evaluated on diffusion-generated content, using frequency-domain and noise-residual cues (the kind of low-level statistical



artifacts diffusion models leave behind in ways visually imperceptible but statistically detectable) rather than face-specific physiological or temporal cues, since fully-synthetic diffusion video may not even contain a consistent “face” to track in the way face-swap forgeries do.

The visual/interactive payoff here is strong: the system produces a heatmap overlay on the input image/frame showing exactly which spatial regions look most “synthetically generated,” directly analogous to Grad-CAM-style visualizations, giving you a live, intuitive, “look, the AI is pointing at the suspicious region” demo that’s immediately legible to a non-technical evaluator.

### 3. Technical Architecture

Data sources: the team needs a diffusion-specific dataset, and this is where the project gets genuinely current rather than recycling FF++; public diffusion-deepfake datasets released in 2024-2025 specifically for this gap (the “Diffusion Deepfake” dataset referenced in the 2024 arXiv literature, plus the team generating their own additional diffusion-fake samples using freely-available open diffusion models, e.g., Stable Diffusion’s video-capable variants or other open-source text-to-video models runnable on a single GPU, which is itself a legitimate and current data-augmentation strategy worth describing explicitly in the report) paired with real video frame datasets (FF++’s real subset, or any general video dataset, as the “real” class).

ML/DL models: a lightweight CNN or hybrid CNN-frequency-domain model (following the 2026 literature’s explicit finding that small, targeted-cue fusion models can outperform large backbones, e.g., a small backbone combined with a wavelet/FFT-derived noise-residual branch) trained for binary real/diffusion-fake classification with a localization head (a segmentation-style output, e.g., a small U-Net-style decoder or a class-activation-map approach) producing the suspicious-region heatmap.

Training pipeline: assemble the real-vs-diffusion-fake frame dataset (public diffusion-fake data plus the team’s own generated samples using open diffusion models); train the classifier with the localization head jointly, using pixel-level or patch-level pseudo-labels where the diffusion generation process is known (e.g., inpainted/edited regions have ground-truth masks from the generation process itself, a genuinely available supervision signal for self-generated training data); explicit data augmentation simulating compression and re-encoding, since real-world diffusion fakes circulating online will have been through several compression passes.

Evaluation pipeline: classification AUC on held-out diffusion-fake test data; localization quality via IoU (intersection-over-union) against the known ground-truth manipulation masks for self-generated test samples; the critical cross-method test, evaluating whether a model trained on one diffusion model’s outputs (e.g., one open-source text-to-video model) generalizes to a different diffusion model’s outputs it never saw during training, which is the single most important and most honestly-reported number in this entire project, directly addressing the 2025 finding that this exact cross-method gap is unresolved in the field.

Deployment architecture: a web dashboard (React/FastAPI) where a user uploads an image or video frame and sees the live heatmap overlay plus a real/fake confidence score, with an additional panel showing the frequency-domain visualization (e.g., the FFT magnitude spectrum) side-by-side with the spatial heatmap, giving a genuinely technical-looking, demo-friendly dual visualization.

## 4. Novelty Analysis

Existing solutions: face-swap-era detectors and rPPG-based methods (including Idea 1) are not designed for fully-synthetic diffusion content and a 2025 paper explicitly states that differential-anomaly methods relying on real-vs-fake frame pairs cannot operate on single completely artificial frames generated by state-of-the-art diffusion methods, while the same paper notes newer text-to-video and image-to-video models like Sora and Runway-Gen3 can create ultra-realistic deepfake videos from just a text prompt or starting frame, a structurally different threat existing tools were not built for. The May 2026 literature on frozen foundation backbones found they remain discriminative for full-face synthesis but reveal fundamental limits when evaluated on localized face edits, suggesting localization specifically (not just classification) is an underexplored, current gap. Your contribution: a lightweight, explicitly diffusion-targeted, localization-capable detector evaluated with a genuine cross-diffusion-model generalization test, a combination not yet saturated in the undergraduate-FYP space (this remains 2025-2026 frontier research territory, genuinely novel for an FYP).

## 5. Dataset Availability

Public datasets: the “Diffusion Deepfake” dataset and similar 2024-released diffusion-forgery benchmarks (the team should search arXiv/Papers-with-Code for the most current public release at project start, since this sub-field is moving fast and newer, larger releases are likely to appear before your project starts); FF++ real-video subset as the “real” class baseline. Data collection strategy: the team generates additional diffusion-fake training/test samples themselves using open-source diffusion video-generation models (several run on a single consumer GPU, particularly at lower resolution/frame-count settings appropriate for a training-data-generation task rather than production-quality output), which is realistic, free, and gives the team direct control over generating cross-method generalization test cases (generate fakes with Model A for training, Model B for testing, the cleanest possible generalization experiment design). Estimated dataset size needed: a few thousand real and diffusion-fake frames/clips is achievable; this project is more bottlenecked by the team’s own generation-compute budget and time than by data availability, since diffusion video generation, even on a single consumer GPU, is slower per-sample than simply downloading an existing dataset, and this should be budgeted for explicitly.

## 6. Development Complexity

Difficulty: Hard, specifically because self-generating training data via diffusion models adds a real compute and engineering overhead beyond just downloading a dataset, and the cross-method generalization design requires careful experiment planning. Estimated completion time: 9-10 months; this is one of the most ambitious ideas on this list and the team should have an explicit fallback (a strong classifier without the localization head) ready by month 6 if the localization component proves harder than expected. GPU requirements: a single consumer GPU (8-12GB) is sufficient for the classifier/localizer training itself; the diffusion-fake-generation step is the more GPU-intensive part of the project and may need to lean on free Colab/Kaggle GPU time budgeted carefully across the team, since generating a few thousand video clips, even short ones, takes real wall-clock time. Team size suitability: best for 3-4



students (one on diffusion-fake data generation, one on the classifier/localizer model, one on the cross-method evaluation design, one on the dashboard).

## 7. Funding Potential

Why investors/incubators/grants may care: this targets the newest, most current threat category in deepfake detection (diffusion/Sora-class fakes), which is exactly the kind of “ahead of the curve” framing that attracts both deep-tech grant funding (Ignite) and is a strong HEC Technology Development Fund fit given its genuine, current research novelty. Potential market size: global and growing rapidly as diffusion-video generation tools become more accessible to the public, meaning the threat (and the market for detecting it) is actively expanding, a strong “growing problem” narrative for any pitch. Commercialization opportunities: API-as-a-service for diffusion-content detection specifically, positioned as a complement to (not replacement for) existing face-swap detectors, a credible niche-but-growing wedge rather than competing head-on with mature deepfake-detection incumbents.

## 8. Research Potential

Very strong, arguably the most frontier-research-adjacent idea on this list. A paper on “cross-method generalization of diffusion-generated video forensics: a localization-based approach” sits at the genuine current edge of the field (2024-2026 literature on this specific sub-problem is still thin, meaning a well-executed undergraduate project could plausibly contribute a citable result, not just a replication); realistic target venues include WIFS, a CVPR/WACV media-forensics workshop, or an ICASSP forensics track. Research questions: does training on one diffusion architecture’s outputs transfer to a structurally different diffusion architecture (e.g., latent-diffusion-based versus a different sampling scheme), and which specific frequency-domain artifacts (if any) are shared across diffusion architectures versus architecture-specific, a genuinely open and valuable empirical question.

## 9. Risks and Challenges

Technical risks: this is the highest-research-risk idea on the list; diffusion-forensic artifacts may be subtler and harder to learn reliably than face-swap artifacts, and the team should be prepared for a genuinely uncertain outcome and report negative/mixed results honestly rather than overstating success, which evaluators will respect more than an inflated claim. Data risks: self-generated training data via diffusion models risks the model learning artifacts specific to the team’s chosen generation settings (resolution, sampler, step-count) rather than diffusion-generation in general; the cross-method test is specifically designed to catch this and must not be skipped. Deployment risks: low; this is a forensic-assistance tool, not a decision-automation system, and should be framed that way throughout.

## 10. Final Verdict

Innovation: 9/10. Practicality: 5/10. Funding Potential: 6/10. Research Potential: 9/10. Ease of Completion: 4/10.

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# IDEA 3: Generalizable Audio Deepfake Detector for Modern Emotional/Expressive TTS (Targeting the 2024-2025 TTS Gap)

## 1. Problem Statement

Audio deepfakes (voice cloning) are used in financial fraud (impersonating a executive or family member in a phone scam), disinformation, and harassment, and a January 2026 paper found that none of the standard, widely-used audio-deepfake detection benchmarks include audio generated by the newest 2024-2025 TTS architectures, meaning detectors trained and evaluated on those benchmarks have an unmeasured, likely-severe blind spot against exactly the newest, most realistic synthetic voices in circulation right now. This affects anyone targeted by voice-cloning fraud (a rapidly growing and well-documented financial-crime category) and platforms/call-centers needing to verify caller authenticity, and matters because the gap is not theoretical, it's a directly measured, currently-unaddressed benchmark blind spot.

## 2. Proposed Solution

An audio deepfake detector explicitly trained and evaluated against audio from modern, expressive TTS systems (not just the older, neutral-speech TTS that existing benchmarks cover), with a specific focus on the emotional/expressive-speech generalization gap the January 2026 paper identifies as nearly unstudied. The system extracts spectral and prosodic features sensitive to the specific artifacts modern neural-codec and non-autoregressive TTS systems leave behind, and is evaluated not just on overall accuracy but specifically on whether performance degrades on emotional/expressive speech versus neutral speech, a genuinely novel evaluation axis.

The visual/interactive payoff: a live dashboard where a user uploads or records audio and sees a real-time spectrogram with the model's attention/saliency overlaid on the specific time-frequency regions driving the fake/real decision, plus a side panel breaking down the decision by detected emotional tone (neutral, emotional/expressive), directly visualizing the exact generalization axis your project is investigating, which is both genuinely informative and demo-compelling.

## 3. Technical Architecture

Data sources: ASVspoof (the standard, well-established academic audio-deepfake benchmark series, multiple generations available, free for academic use), Deepfake-Eval-2024's audio component if available, and critically, the team generating their own modern-TTS-sourced fake-audio samples using freely accessible, open or freemium modern TTS systems (several open-source neural TTS systems with expressive/emotional speech capability are runnable on a single consumer GPU or even CPU for inference, since TTS inference is far less compute-intensive than training), explicitly targeting the emotional-speech gap the literature identifies.

ML/DL models: a spectral feature extractor (log-mel spectrogram or a wav2vec2-style self-supervised audio embedding, both standard, well-documented, and modest in compute requirement for inference/fine-tuning) feeding a lightweight classifier (a small CNN or a shallow Transformer encoder, following the same "small targeted model" lesson from the 2026

visual-deepfake literature, which plausibly transfers to the audio domain as a sound methodological choice to state explicitly); critically, the model is trained with an explicit emotional/expressive-speech-aware data split, ensuring both training and evaluation include a meaningful proportion of non-neutral speech, which is the project's specific contribution given that most prior work has not done this systematically.

Training pipeline: combine ASVspoof's real/fake audio with the team's own modern-TTS-generated samples (both neutral and emotional/expressive prompts, explicitly tagged by emotional category at generation time, since the team controls the generation and knows the ground-truth emotional intent); train the classifier with standard cross-entropy loss; deliberately construct a stratified evaluation split that separates neutral-speech and emotional-speech test performance, rather than reporting one pooled accuracy number that could hide the gap.

Evaluation pipeline: standard Equal Error Rate (EER, the conventional audio-spoof-detection metric, more standard in this literature than plain accuracy) on the ASVspoof held-out test set (the "did we replicate known results" baseline); EER specifically broken out by neutral versus emotional/expressive speech on the team's own modern-TTS test set (the project's central, novel empirical contribution); a cross-TTS-system generalization test (train detection on samples from TTS System A, test on TTS System B never seen during training), mirroring the same generalization-rigor pattern as Ideas 1 and 2 above.

Deployment architecture: a web app (React/FastAPI) with live audio upload/recording, the spectrogram-plus-saliency visualization described above, and a results panel showing the real/fake verdict, confidence, and the detected-emotional-category breakdown; this could also be extended (stretch goal) to a simple phone-call-style demo scenario for the final viva, e.g., "is this voicemail real or cloned," which is a highly relatable, immediately understandable demo for a non-technical evaluator.

## 4. Novelty Analysis

Existing solutions: ASVspoof and similar benchmarks are well-established and heavily used, but the January 2026 paper is explicit that none of ASVspoof 2021, Deepfake-in-the-Wild, or Deepfake-Eval-2024 include audio from the latest TTS architectures released in 2024-2025, which the paper characterizes as not a minor oversight but a critical vulnerability in the field's collective understanding of modern deepfake threats, and further notes that emotional and non-neutral speech, despite being increasingly common in modern TTS output, has received surprisingly little research attention as a generalization dimension. Your contribution: being among the first systematic evaluations to explicitly construct and report an emotional-versus-neutral-speech generalization breakdown using current-generation TTS systems, a gap the literature itself flags as open right now, meaning this idea has unusually fresh, citable novelty for an undergraduate project.

## 5. Dataset Availability

Public datasets: ASVspoof (multiple released editions, standard academic-use access, widely used and well-documented, low access friction). Data collection strategy: the team generates the modern-TTS and emotional-speech samples themselves using accessible open or freemium

TTS tools, which is the project's main data-engineering task and should be budgeted real time (designing a balanced set of prompts across emotional categories, generating audio, and tagging ground truth at generation time, all legitimate, describable methodology). Estimated dataset size needed: a few thousand audio clips combining ASVspoof's real/fake data and the team's own generated modern-TTS samples (split across neutral and several emotional categories) is realistic and sufficient, since audio classification tasks generally need less raw data volume than video tasks to reach a reasonably informative result.

## 6. Development Complexity

Difficulty: Medium. Estimated completion time: 8 months. GPU requirements: low; spectral-feature-based audio classifiers are small and train quickly on a single consumer GPU or even CPU; TTS generation for the dataset-construction phase is inference-only (not training a TTS model from scratch) and similarly modest in compute need. Team size suitability: 3 students (one on the TTS-sourced dataset construction, one on the classifier and cross-system evaluation, one on the dashboard/visualization).

## 7. Funding Potential

Why investors/incubators/grants may care: voice-cloning fraud is a fast-growing, well-publicized financial-crime category internationally, making audio-authenticity verification a genuinely commercially relevant security category; this is a strong fit for cybersecurity-adjacent deep-tech grant funding (Ignite) and is positionable to security-focused incubators/accelerators globally given the universality of the threat (not Pakistan-specific). Potential market size: large and global; call centers, banks' phone-verification systems (note: this is about voice-authenticity infrastructure, not the excluded "bank credit scoring" category, so it remains within your constraints), and consumer-facing scam-detection apps are all plausible markets. Commercialization opportunities: an API-as-a-service for voice-authenticity verification, or a consumer-facing browser/phone-app demo ("check if this voicemail/call is a clone"), which is an unusually relatable, easy-to-explain consumer pitch for a hackathon-style or incubator demo day.

## 8. Research Potential

Strong. A paper on "generalization of audio deepfake detection to modern expressive text-to-speech: an emotional-speech evaluation" is directly publishable, sitting exactly inside a gap the field's own most recent literature flags as open; realistic target venues include Interspeech, ICASSP's audio-forensics track, or the ASVspoof challenge's own associated workshop (which explicitly invites exactly this kind of generalization-focused contribution). Research questions: does detection performance degrade more on emotional/expressive speech than neutral speech, and if so, by how much and on which specific emotional categories (a clean, fundable, currently-unanswered empirical question); does a detector trained on one modern TTS system's output transfer to a structurally different TTS architecture (autoregressive versus non-autoregressive, or different neural-codec designs).

## 9. Risks and Challenges

Technical risks: modern open TTS systems vary significantly in quality and accessibility, and the team should validate early (month 1) that they can reliably generate the emotional/expressive audio samples needed before committing fully to this specific angle. Data risks: self-generated TTS data, as with Idea 2's diffusion-video data, risks the model learning artifacts specific to the team's chosen TTS tools rather than modern TTS generally; the cross-TTS-system generalization test exists specifically to catch and honestly report this limitation. Deployment risks: low-to-moderate; a wrongly-cleared cloned-voice fraud call has real victim impact, so the tool should be clearly positioned as a verification aid, not a guarantee, in both the interface and any public-facing description.

## 10. Final Verdict

Innovation: 9/10. Practicality: 7/10. Funding Potential: 7/10. Research Potential: 9/10. Ease of Completion: 7/10.

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## IDEA 4: Real-Time Multi-Agent Crypto Market Microstructure Visualizer (Non-Banking FinTech)

### 1. Problem Statement

Retail crypto traders and analysts have access to price charts but very little visibility into the underlying market microstructure dynamics driving short-term price behavior (order-flow imbalance, whale-wallet movement patterns, exchange-to-exchange arbitrage pressure, social-sentiment-driven volume spikes), and existing retail tools either show raw, hard-to-interpret order-book data or oversimplified price charts with no behavioral-modeling layer connecting the two. This affects retail crypto traders, market researchers, and crypto-analytics-curious developers, and matters because genuinely understanding *why* a market is moving (not just that it moved) is the actual value proposition that distinguishes a serious analytics tool from a basic price-chart app.

### 2. Proposed Solution

A multi-agent simulation-plus-real-data system where several specialized “analyst agents” each independently process a different data stream (on-chain whale-wallet movements via public blockchain data, exchange order-book imbalance via public exchange APIs, and social-sentiment signal via public social-media-adjacent text sources), and a coordinating layer synthesizes these into a live, visual “market pressure” dashboard, explicitly showing which signal (whale movement, order-flow imbalance, sentiment) is currently dominant, with each agent's individual reasoning visible and inspectable, not hidden behind a single aggregate score.

This is visually and interactively excellent by design: imagine a live dashboard with three animated “pressure gauges” (one per agent/signal type) feeding into a central combined-pressure indicator, plus a live order-book heatmap and an on-chain whale-transaction live feed,

giving a viva audience an immediately legible, genuinely impressive “this looks like a real trading-floor tool” demo.

### 3. Technical Architecture

Data sources: public on-chain data via free blockchain explorers’ APIs (e.g., Etherscan-style APIs, free tier, for whale-wallet-movement tracking on a major chain), public cryptocurrency exchange APIs (e.g., Binance’s public market-data API, free, no special access needed, for live order-book and trade-flow data), and public social-text sources for sentiment (e.g., public crypto-subreddit data or similarly public text sources, scraped within platform terms-of-service limits, not private data).

ML/DL models: a sentiment-classification model (a fine-tuned lightweight transformer, e.g., a DistilBERT-class model fine-tuned on a crypto/finance-specific sentiment dataset, modest compute) for the social-signal agent; a time-series anomaly/imbalance-detection model (a relatively simple statistical or lightweight ML model detecting order-flow imbalance patterns, e.g., volume-weighted imbalance metrics combined with a small gradient-boosted classifier predicting short-term directional pressure) for the order-book agent; a rule-based-plus-clustering approach (e.g., clustering wallet addresses by transaction-size and frequency patterns to flag likely “whale” wallets, a legitimate unsupervised-learning task) for the on-chain agent; a lightweight coordinating model or rule-based fusion layer combining the three agents’ signals into the unified pressure indicator, with the fusion logic itself being interpretable and visualizable (e.g., a simple weighted-combination or small learned-fusion model, deliberately kept transparent rather than an opaque deep fusion network, since interpretability is part of the product’s value proposition here).

Training pipeline: fine-tune the sentiment model on a public crypto/finance-sentiment-labeled dataset (several exist on Kaggle and via NLP-finance research, detailed below); fit the order-flow-imbalance model on historical exchange order-book/trade data with backtested directional-pressure labels (e.g., did price move up/down significantly in the next N minutes following a given imbalance pattern, a self-supervised labeling approach requiring no manual annotation); the whale-detection clustering is unsupervised and validated qualitatively against known large-wallet behavior patterns described in on-chain-analytics literature/community resources.

Evaluation pipeline: sentiment-model standard classification metrics (F1) against its labeled test set; order-flow-imbalance model evaluated via its directional-prediction accuracy/AUC on historical held-out time periods (chronological split, never random, since this is time-series data exactly as in the FinTech Volume 1 report’s trading-backtester caveat); the overall system evaluated qualitatively and via a small retrospective case-study analysis (e.g., “did the combined pressure indicator show elevated signal ahead of three known historical price moves the team selects as case studies”), which is an honest, appropriately-scoped evaluation approach for a multi-agent system where ground truth for “correct market interpretation” doesn’t cleanly exist.

Deployment architecture: a real-time web dashboard (React frontend with WebSocket live updates, FastAPI backend orchestrating the three agents) displaying the three live pressure gauges, the order-book heatmap, the on-chain whale-feed, and the combined-pressure timeline,



all updating live as new exchange/blockchain data arrives, the single most demo-compelling deployment architecture on this list given the requirement for visual, real-time, “watch the AI work” interactivity.

#### 4. Novelty Analysis

Existing solutions: crypto-analytics platforms exist (some free, many paid/institutional) and do provide on-chain and order-book data, but they typically present raw data dashboards (charts and numbers) rather than a multi-agent, explainable, synthesized “why is the market moving right now” interpretation layer; sentiment-analysis-for-crypto tools exist separately but are rarely fused transparently with on-chain and order-flow signals in one interpretable system. Your contribution: the explicit multi-agent architecture with each agent’s individual reasoning kept visible and inspectable (rather than fused into an opaque single score), aimed specifically at the explainability and “watch the AI reason” interactivity gap in existing retail crypto-analytics tools, which is a genuinely different product framing from both raw-data dashboards and opaque black-box signal services.

#### 5. Dataset Availability

Public datasets: free-tier blockchain-explorer APIs (no special access barriers, standard for on-chain data), free public cryptocurrency-exchange market-data APIs (similarly standard, no special access barriers for the historical/live order-book and trade data needed), and publicly available crypto/finance-sentiment-labeled datasets on Kaggle (several exist, typically a few thousand to tens of thousands of labeled social-text samples) for the sentiment-model fine-tuning. Data collection strategy: primarily API-pull for on-chain and exchange data (low-risk, well-documented, standard practice); social-sentiment training data comes from existing public labeled datasets rather than the team needing to scrape and label from scratch, reducing project risk. Estimated dataset size needed: a few thousand labeled sentiment samples for fine-tuning (achievable from existing public datasets), plus several months of historical exchange/on-chain data (trivially available via free API history) for the order-flow and whale-detection components.

#### 6. Development Complexity

Difficulty: Medium (the individual components are each well-precedented and tractable; the main complexity is the real-time multi-agent orchestration and live-dashboard engineering, which is genuine but manageable software-engineering work, not deep research risk). Estimated completion time: 8 months. GPU requirements: low; the sentiment-model fine-tuning is the only GPU-relevant component and is modest (a DistilBERT-class model on a few thousand samples trains quickly on a single consumer GPU or free Colab/Kaggle GPU); the order-flow and on-chain components are not GPU-bound. Team size suitability: 3-4 students (one per agent: sentiment, order-flow, on-chain, plus one on the real-time dashboard/orchestration layer, or the dashboard work distributed across the team if only 3 members).

## 7. Funding Potential

Why investors/incubators/grants may care: crypto-analytics and market-intelligence tools are an established, internationally fundable fintech category (explicitly within your “modern finance” framing, not the excluded banking category), and the explainable, multi-signal-fusion angle is a genuine differentiator investors in this space look for given how saturated raw-data crypto-dashboard tools already are. Potential market size: global; retail crypto traders are a large, internationally distributed user base, and the product requires no region-specific licensing or regulatory entanglement (crucially, the project does not touch actual trade execution, only analytics/visualization, simplifying both the FYP scope and any future commercialization path). Commercialization opportunities: freemium SaaS (free basic dashboard, paid tier for more assets/alerts/historical depth) is the standard, proven model in this exact product category, giving the team credible market comparables.

## 8. Research Potential

Moderate. A paper on “multi-agent, explainable market-pressure synthesis for cryptocurrency microstructure analysis” is a reasonable, if less academically-conventional, contribution; the more research-credible framing is narrower, e.g., “does order-flow-imbalance signal Granger-cause short-term sentiment shifts (or vice versa) in cryptocurrency markets,” a specific, testable empirical-finance question the project’s data pipeline could directly support as a side analysis, giving the team a more traditional, defensible research angle alongside the product-demo angle. Research questions: which of the three signal types (on-chain, order-flow, sentiment) is most predictive of near-term price moves, and does that ranking change across different market regimes (calm versus volatile), a genuinely interesting empirical question the team’s own data pipeline is well-suited to answer.

## 9. Risks and Challenges

Technical risks: real-time multi-API orchestration (exchange API, blockchain API, and the model-inference pipeline all running live) has real engineering failure modes (API rate limits, downtime, data-format changes) that must be handled gracefully, not assumed away; the team should budget real integration-testing time, not just model-development time. Data risks: free-tier API rate limits can constrain how much live historical backtesting the team can do; caching strategy should be planned early. Deployment risks: low; this is explicitly an analytics/visualization tool, not a trade-execution or investment-advice system, and the dashboard should state this clearly (no “buy/sell” recommendations, only signal visualization) both to stay safely scoped and to avoid any unintended financial-advice framing.

## 10. Final Verdict

Innovation: 7/10. Practicality: 8/10. Funding Potential: 7/10. Research Potential: 5/10. Ease of Completion: 8/10.

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# IDEA 5: Agentic Personal Expense-Intelligence System with Behavioral Spending-Pattern Simulation (Non-Banking FinTech)

## 1. Problem Statement

People struggle to understand their own spending behavior beyond simple category totals (e.g., “you spent \$400 on food this month”), because existing personal-finance apps mostly categorize and total transactions without modeling the underlying behavioral patterns (impulse-spending triggers, day-of-week/mood-adjacent spending spikes, subscription creep) that actually drive financial stress. This affects individual consumers trying to build healthier financial habits, and matters because category totals alone are well-documented in behavioral-finance research as insufficient to actually change spending behavior, people need to see the *pattern*, not just the total, to act on it.

## 2. Proposed Solution

This is explicitly not a recommendation system or basic categorization app (per your exclusion), it’s an agentic behavioral-pattern-discovery and simulation system: an agent analyzes a user’s transaction history to discover latent spending-behavior patterns (e.g., “spending spikes 40% on Friday evenings,” “three small subscription charges have crept in over six months without the user noticing,” “grocery spending shows a recurring late-month spike consistent with running out of pre-planned meals”), then runs a forward-looking simulation showing how specific behavior changes (canceling a subscription, capping a category) would have changed historical outcomes, letting the user interactively explore “what if” scenarios rather than receiving a static report.

This is visually excellent: an interactive timeline/calendar heatmap of spending intensity, a pattern-discovery panel where each discovered pattern is presented with its supporting evidence (the specific transactions and dates that revealed it), and a live “what-if” simulator slider letting the user adjust a behavior and watch a simulated alternate-history spending curve redraw in real time, genuinely interactive and satisfying to demo.

## 3. Technical Architecture

Data sources: public personal-finance transaction datasets for model development and demo purposes (several exist, detailed below), plus the team’s own (consenting, anonymized) transaction exports as realistic test cases, and critically, a synthetic transaction-generator the team builds to create realistic, pattern-rich synthetic users for demo purposes (since real transaction data is sensitive and hard to obtain at scale, and a well-designed synthetic generator is itself a legitimate, describable engineering contribution, not a shortcut).

ML/DL models: an unsupervised pattern-discovery layer combining time-series decomposition (separating trend/seasonality/residual in spending-by-category time series, a classical and well-documented technique) with clustering (e.g., discovering recurring temporal spending clusters via DBSCAN or a similar density-based method, well-suited to irregular real-world transaction timing) and a change-point-detection algorithm (e.g., for catching the “subscription crept in” pattern, where a small recurring charge begins partway through the observed history,

a genuinely useful and not-overly-complex classical-statistics technique); an LLM-based agent layer (a small local quantized model, or sparing hosted-API calls) that takes the discovered statistical patterns and generates the natural-language explanation and supporting-evidence narrative, explicitly grounded in the actual discovered pattern data (retrieval-augmented-generation style, citing specific transactions, not generating ungrounded financial advice); a lightweight simulation engine (a deterministic or simple probabilistic forward-projection model, not a deep generative model, since interpretability matters more than generative sophistication here) for the interactive what-if exploration.

Training pipeline: the pattern-discovery layer is primarily unsupervised/classical-statistical and does not require a large labeled training set, it's fit per-user on their own transaction history at inference time (a genuinely different and appropriate ML paradigm choice from the supervised-classification pattern in most other ideas on this list, worth explicitly explaining and justifying in the report); the LLM-agent layer is engineered (prompt design, retrieval grounding) rather than fine-tuned, an appropriately-scoped agentic-AI architecture choice for the timeline.

Evaluation pipeline: pattern-discovery quality evaluated against the synthetic data generator's known ground-truth injected patterns (since the team controls what patterns were synthetically injected, they can directly measure precision/recall of pattern discovery, a clean and honest evaluation design); a qualitative user-study component (even a small one, 5-10 willing participants reviewing the system's analysis of their own anonymized data, or reviewing realistic synthetic personas) assessing whether the discovered patterns and explanations are perceived as genuinely insightful, which is the most meaningful evaluation for this kind of behavioral-insight product.

Deployment architecture: a web app (React/FastAPI) with the interactive calendar-heatmap, pattern-discovery panel with evidence drill-down, and the live what-if simulator slider described above; transaction data handled with explicit, visible privacy framing (processed locally/session-based rather than persisted, a deliberate design and ethical choice worth highlighting to evaluators given the sensitivity of financial data).

#### 4. Novelty Analysis

Existing solutions: mainstream personal-finance apps (Mint-style category trackers, and similar) provide categorization and totals but rarely surface behavioral patterns with evidence-grounded explanations, and essentially none provide an interactive, simulation-based "what if I changed this behavior" exploration tool; this is explicitly distinct from a recommendation system (which would just say "spend less on X") because it surfaces the *pattern* and lets the user explore consequences themselves, a meaningfully different and more behaviorally-grounded product design. Your contribution: combining classical change-point/pattern-discovery statistics with an LLM-based, evidence-grounded explanation layer and an interactive forward-simulation tool, a combination not common in either the academic personal-finance-AI literature or existing commercial tools, which tend to do one or the other but not the full pipeline together.

## 5. Dataset Availability

Public datasets: several public personal-finance/transaction-categorization datasets exist on Kaggle (typically a few thousand to tens of thousands of labeled or semi-labeled transaction records, suitable for validating the categorization and pattern-discovery pipeline) and academic personal-finance-research-adjacent datasets occasionally released alongside behavioral-finance papers. Data collection strategy: the team's synthetic transaction-generator (built explicitly to inject known, controllable behavioral patterns) is the primary tool for both development and evaluation, supplemented by the public datasets for realism-calibration (ensuring the synthetic data's statistical properties resemble real spending data) and by the team's own voluntarily-shared, anonymized transaction history as a small real-world validation case. Estimated dataset size needed: synthetic generation can produce as much data as needed (hundreds of synthetic user-histories with injected patterns is easily achievable); the public Kaggle datasets at a few-thousand-record scale are sufficient for the realism-calibration step.

## 6. Development Complexity

Difficulty: Medium. Estimated completion time: 8 months. GPU requirements: very low; the core pattern-discovery pipeline is classical statistics/ML, not deep learning, and the only GPU-relevant component (if using a local LLM rather than a hosted API for the explanation-generation step) is a small quantized model running comfortably on a single consumer GPU with modest VRAM. Team size suitability: 3 students (one on the pattern-discovery statistical pipeline and synthetic-data generator, one on the LLM-agent explanation/simulation layer, one on the interactive dashboard).

## 7. Funding Potential

Why investors/incubators/grants may care: personal financial-wellness tools are a genuinely large and currently active consumer-fintech category internationally (distinct from and not overlapping with the excluded banking/credit-scoring category, since this never touches lending or credit decisions at all), and the explainable, simulation-based, non-recommendation-engine framing is a credible differentiator in a market that investors increasingly view as saturated with generic budgeting apps. Potential market size: very large, global, consumer-direct (every individual with bank-linked transaction data is a potential user), though monetization in consumer fintech is famously harder than B2B (a fair caveat to state plainly rather than oversell). Commercialization opportunities: freemium consumer app (most realistic path), or a white-label/B2B2C licensing model selling the pattern-discovery engine to existing personal-finance apps or neobanks wanting to add this capability without building it themselves, which is arguably the more realistic near-term commercialization path for a small founding team without consumer-app distribution resources.

## 8. Research Potential

Moderate. A paper on "evidence-grounded behavioral pattern discovery and counterfactual simulation for personal financial data" is a reasonable contribution at the intersection of applied ML and behavioral/computational finance, potentially fitting a fintech-AI workshop (ICAIF, again, given its breadth) or a human-computer-interaction-adjacent venue if the user-study component is developed further, since the interactive-simulation/explanation-quality

angle has a genuine HCI dimension worth emphasizing if the team wants to pursue that specific research framing. Research questions: does evidence-grounded natural-language explanation of discovered spending patterns improve user-reported insight/actionability compared to a no-explanation, statistics-only baseline (a clean, human-subjects-evaluable comparison); how reliably does the change-point-detection approach catch synthetically-injected “subscription creep” patterns across varying noise levels in the synthetic data, a clean, fully-controllable technical evaluation.

## 9. Risks and Challenges

Technical risks: real personal transaction data is genuinely messy (inconsistent merchant-name formatting, ambiguous categorization) in ways synthetic data may not fully capture; the team should validate against at least some real (consenting, anonymized) data, not rely on synthetic data alone, to keep the project’s claims honest. Data risks: privacy is the central concern here, even with anonymization; the team should be explicit and conservative about data handling (session-based processing, no persistent storage of real user financial data, clear consent for any real-data testing) both as good practice and because evaluators will specifically look for this given the sensitivity of the domain. Deployment risks: an agent that generates an incorrect or financially-unwise “what if” simulation could mislead a user about their finances; the simulation should be clearly labeled as an illustrative projection based on historical patterns, not a financial-advice guarantee, both in the UI and the final report’s ethical considerations section.

## 10. Final Verdict

Innovation: 7/10. Practicality: 7/10. Funding Potential: 6/10. Research Potential: 5/10. Ease of Completion: 7/10.

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# IDEA 6: Synthetic Media Provenance Chain Visualizer for Image Editing History (Media Forensics)

## 1. Problem Statement

Beyond full-face deepfakes, a related and increasingly common problem is incremental, partial image manipulation (selective object insertion/removal, localized AI-inpainting edits, splicing), and existing detection tools mostly give a single binary “manipulated/not” verdict without showing *what* was edited, *how many* edit operations likely occurred, or in what *plausible order*, information that matters enormously to journalists and forensic investigators trying to understand an image’s actual history, not just flag it as suspicious. This affects photojournalism verification teams, forensic investigators, and fact-checking organizations, and matters because a binary verdict is far less actionable than a reconstructed edit history when the goal is actual investigation, not just filtering.



## 2. Proposed Solution

A system that detects multiple distinct manipulated regions within a single image (not just one global verdict), classifies the likely manipulation type per region (splice/copy-move, AI-inpainting, color/lighting adjustment), and presents a visual “provenance chain” reconstruction, an ordered, illustrated hypothesis of which edits likely happened and in roughly what sequence, based on boundary-artifact and noise-residual cues at each detected region’s edges. This is explicitly not a single classifier, it’s a multi-stage analysis pipeline producing a genuinely investigative output.

The visual payoff is strong and specific to forensic/investigative demos: the dashboard shows the uploaded image with multiple highlighted regions, each clickable to reveal its specific manipulation-type classification and confidence, plus a simple “likely edit sequence” timeline visualization, which is a substantively different and more sophisticated-looking demo than a single fake/real percentage.

## 3. Technical Architecture

Data sources: established image-forensics datasets with region-level manipulation ground truth (e.g., datasets from the image-forensics research community that provide pixel-level manipulation masks, including splicing and copy-move benchmark datasets, and CASIA-style tampering datasets, all standard and well-documented in academic image-forensics literature), supplemented by the team generating additional AI-inpainting-edited samples using open image-inpainting models (e.g., Stable Diffusion’s inpainting variants, runnable on a single consumer GPU), since AI-inpainting-specific manipulation data is the newer, less-covered category in older splice/copy-move-focused academic datasets.

ML/DL models: a multi-region manipulation-localization model (a segmentation-style architecture, e.g., a U-Net or a lightweight segmentation transformer, trained to output a per-pixel manipulation-likelihood map, building on the same “small, targeted model” lesson from the broader 2026 literature rather than reaching for an oversized backbone) producing candidate manipulated regions; a per-region manipulation-type classifier (a smaller CNN operating on cropped candidate regions) distinguishing splice/copy-move from AI-inpainting from simple color/lighting adjustment, since these manipulation types leave statistically different residual signatures; a boundary-artifact-based heuristic-plus-learned-ranking layer for the provenance-ordering hypothesis (explicitly described and evaluated as a heuristic-assisted hypothesis, not a guaranteed-correct sequence, an honest framing the report should state clearly).

Training pipeline: train the localization segmentation model on the combined academic splice/copy-move datasets plus the team’s self-generated AI-inpainting samples (with known ground-truth masks from the generation process, the same legitimate self-supervision strategy used in Idea 2); train the per-region manipulation-type classifier on cropped, labeled regions from the same combined dataset; the provenance-ordering component is primarily rule-based/heuristic (using boundary-sharpness and compression-artifact cues as ordering signals) rather than a fully learned model, an appropriately-scoped choice given the genuine difficulty of the ordering sub-problem.

Evaluation pipeline: standard localization metrics (pixel-level F1/IoU against ground-truth manipulation masks) for the segmentation model; per-region classification accuracy for the manipulation-type classifier; a qualitative case-study evaluation for the provenance-ordering hypothesis (since rigorous quantitative ground truth for “edit sequence” is genuinely hard to obtain even in a self-generated dataset unless the team carefully designs synthetic multi-step-edited images with known ground-truth edit order, which is achievable and worth doing specifically to get at least some quantitative ordering-accuracy number, not purely qualitative).

Deployment architecture: a web app (React/FastAPI) with the multi-region highlighted-image display, click-to-reveal per-region classification, and the edit-sequence-timeline visualization described above, aimed and framed explicitly as an investigative-assistance tool for forensic analysts/journalists, not an automated verdict system.

#### 4. Novelty Analysis

Existing solutions: image-forensics research has long studied splice/copy-move detection (a mature sub-field with established benchmarks) and, more recently, AI-inpainting detection has emerged as its own active area, but the two are rarely combined into one multi-region, multi-manipulation-type system with an explicit provenance-ordering hypothesis layer; most existing tools (academic or commercial) give a single localization map or a single verdict, not a reconstructed, ordered, multi-region edit narrative. Your contribution: combining classic splice/copy-move detection with modern AI-inpainting detection in one unified multi-region pipeline, plus the provenance-ordering hypothesis layer, which, even framed honestly as heuristic-assisted rather than rigorously proven, is a genuinely more investigation-useful output format than existing single-verdict tools provide.

#### 5. Dataset Availability

Public datasets: established academic image-tampering/splicing datasets with region-level ground-truth masks (standard, well-documented, multiple options exist in the image-forensics research literature, typically free for academic use with straightforward access). Data collection strategy: supplement with team-generated AI-inpainting-edited images (using open inpainting models, with full control over ground-truth masks and, importantly, full control over constructing multi-step-edited images with known ground-truth edit order for the provenance-ordering evaluation specifically). Estimated dataset size needed: a few thousand images combining the academic splice/copy-move datasets and the team’s self-generated AI-inpainting and multi-step-edit samples is achievable and sufficient for this scale of project.

#### 6. Development Complexity

Difficulty: Medium-Hard (the localization and per-region classification components are tractable and well-precedented; the provenance-ordering component is the genuinely novel and harder part, and should be scoped as a clearly-labeled exploratory contribution, not oversold as a solved problem). Estimated completion time: 8-9 months. GPU requirements: a single consumer GPU (8GB+) is sufficient for both the segmentation model and the per-region classifier; the inpainting-based synthetic-data generation step has a similar modest compute profile to Idea 2’s diffusion-video generation, but for static images, which is meaningfully less compute-intensive than video generation. Team size suitability: 3-4 students (one on the

localization/segmentation model, one on the per-region classifier, one on the provenance-ordering heuristic and synthetic multi-step-edit dataset construction, one on the dashboard).

## 7. Funding Potential

Why investors/incubators/grants may care: image-forensics-for-journalism/investigation tooling fits both deep-tech innovation grants (Ignite) and civic-tech/media-integrity grant funding (international press-freedom and digital-rights funders, similar to the misinformation-tooling funding landscape noted in prior FYP analyses for this team), though, similar to other media-forensics-for-journalism ideas, this is more grant/NGO-fundable in the near term than a fast-revenue VC pitch, a distinction worth stating honestly. Potential market size: a real but specialized market, forensic investigators, fact-checking organizations, journalism-integrity teams, smaller than a mass-consumer product but with genuine institutional buyers. Commercialization opportunities: licensed as an investigative tool to fact-checking organizations and media-forensics teams, or positioned as a feature/module within a broader media-authentication product suite (a credible integration angle if the team's other deepfake-detection ideas, e.g., Idea 1 or 2, are pursued by a related team, since these could plausibly be packaged together into one "media authenticity suite" pitch).

## 8. Research Potential

Good. A paper on "multi-region, multi-type manipulation localization with provenance-ordering hypothesis generation for image forensics" fits the active image-forensics research community (venues like WIFS, ICASSP's forensics track, or an IH&MMSec-style venue, the dedicated information-hiding/multimedia-security conference, are realistic, undergraduate-reachable targets). Research questions: how accurately can boundary-artifact and compression-residual cues recover the correct edit order in synthetically-constructed, multi-step-edited test images (a clean, fully-controllable quantitative question); does combining classic splice/copy-move detection with modern AI-inpainting detection in one unified model outperform two separately-trained specialist models, a genuinely useful and directly-testable architectural question.

## 9. Risks and Challenges

Technical risks: the provenance-ordering hypothesis is the riskiest component and may not reach high accuracy; this should be explicitly scoped and reported as exploratory/heuristic, with the localization-plus-classification components serving as the more rigorously-evaluated core contribution and a safe fallback if ordering proves too hard. Data risks: minimal beyond the standard self-generated-data caveats already discussed for similar ideas on this list. Deployment risks: a wrong provenance-ordering hypothesis presented too confidently could mislead an actual journalistic investigation; the UI must clearly communicate confidence levels and the heuristic, hypothesis-generating (not definitive) nature of the ordering output.

## 10. Final Verdict

Innovation: 7/10. Practicality: 6/10. Funding Potential: 5/10. Research Potential: 7/10. Ease of Completion: 6/10.

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## IDEA 7: Agent-Based Market Regime Stress-Test Sandbox with Live Visual Simulation (Non-Banking FinTech)

### 1. Problem Statement

Retail investors and finance students have no accessible, visual, hands-on way to understand how different market participant behaviors (panic-selling herds, momentum-chasing traders, contrarian value-buyers) interact to produce emergent market phenomena like flash crashes, bubbles, and recoveries, because this kind of agent-based market simulation has historically lived in expensive institutional risk-modeling software or dense academic papers with no visual, explorable interface accessible to a student or retail audience. This affects retail investors and finance students wanting genuine intuition (not just a textbook description) of *why* markets crash and recover the way they do, and matters because agent-based market modeling is a well-established but historically inaccessible research technique that an accessible visual tool could genuinely democratize.

### 2. Proposed Solution

An interactive agent-based market simulation where the user can populate a simulated market with configurable populations of trader-agent archetypes (momentum-chasers, contrarians, panic-sellers, fundamental-value-investors, each with ML-learned or parameterized behavioral rules calibrated against real historical market-behavior patterns), run the simulation, and watch emergent price dynamics unfold visually in real time, then compare the simulation's emergent statistical properties (volatility clustering, fat-tailed returns, the same "stylized facts" framework used in serious empirical finance) against real historical market data to validate how realistically the simulation behaves.

This is the single most visually and interactively compelling idea on this entire list: a live, animated simulation where you watch agent populations interact and a price chart emerge in real time from their interactions, with sliders letting the user adjust population mix (e.g., "what happens if 30% of the market is panic-prone") and immediately re-run the simulation, giving an immediately understandable, viscerally engaging "watch the AI agents create market behavior" demo perfectly suited to a viva audience, including non-technical evaluators.

### 3. Technical Architecture

Data sources: free historical OHLCV market data (yfinance, Alpha Vantage free tier, the same accessible sources used in conventional quant-finance FYP work) serves as the real-data calibration and validation target, not as training data for a supervised model in the conventional sense, since this project's core ML contribution is calibrating agent behavioral parameters to reproduce realistic emergent statistics, a meaningfully different and more creative ML framing than a standard classifier/predictor.

ML/DL models: rule-based or lightly-parameterized agent archetypes (each archetype's trading-decision rule is a relatively simple function of recent price history and the agent's

behavioral-bias parameters, e.g., a momentum-chaser's buy/sell threshold as a function of recent price trend, a contrarian's threshold inverted) form the simulation's core, deliberately interpretable rather than opaque deep-learned policies, since interpretability is central to the product's educational value; a calibration layer using either a genetic algorithm or a simpler grid/Bayesian-optimization search (e.g., via Optuna, a well-documented, free, easy-to-use hyperparameter-optimization library) tunes the agent-population mix and behavioral parameters to minimize the statistical distance between the simulation's emergent return-distribution properties and a real historical market's actual stylized facts (volatility clustering, fat tails, autocorrelation structure, directly reusing the same empirical-finance evaluation framework as the Volume 1 report's synthetic-time-series-generator idea, a methodologically consistent and well-grounded choice if your evaluator has seen that framing before).

Training pipeline: this project's "training" is better described as calibration: run the agent-based simulation repeatedly across the optimization search, scoring each run against how closely its emergent statistical properties match a real target market period's actual statistics, and converge on the population-mix and parameter settings that best reproduce real market behavior; this is a legitimate, if less conventional, ML pipeline (simulation-based calibration/optimization rather than classification/regression) and should be explicitly described and justified as such in the report, since an evaluator unfamiliar with agent-based modeling may initially expect a more standard supervised pipeline.

Evaluation pipeline: the same stylized-facts framework (volatility clustering via autocorrelation of squared returns, excess kurtosis for fat tails, near-zero raw-return autocorrelation) used to evaluate how well the calibrated simulation reproduces real market statistical behavior, directly analogous to and reusable from the evaluation design of any team also pursuing the synthetic-financial-data-generator idea from prior FYP analysis for this same team, giving a nice methodological consistency story if both projects exist in the same department/cohort; additionally, qualitative case-study validation (does the calibrated simulation, when shocked with a sudden negative price event, produce a flash-crash-and-recovery pattern visually and statistically consistent with real historical flash-crash episodes).

Deployment architecture: a real-time interactive web app (this strongly benefits from a framework supporting smooth live animation, e.g., a React frontend with a canvas/WebGL-based visualization layer, or a Python-based interactive framework like a custom-built Streamlit-plus-Plotly real-time setup if the team wants faster development at some cost to animation smoothness) where the user configures the agent population via sliders, triggers a simulation run, and watches the price chart and agent-population visualization animate live, with a comparison panel showing the simulation's stylized-facts statistics side-by-side with a real historical reference market.

#### 4. Novelty Analysis

Existing solutions: agent-based market modeling is an established academic research technique (with a real, citable literature going back decades in computational/behavioral finance) but almost no accessible, free, visually-interactive consumer or student-facing tool exists for it; what exists is either dense academic simulation code with no interface, or expensive institutional risk-modeling software entirely inaccessible to students/retail users. Your contribution: making agent-based market simulation genuinely accessible and visually

explorable for the first time to a non-institutional audience, calibrated transparently against real market stylized facts rather than presented as an arbitrary toy simulation, a combination of accessibility and empirical rigor that's genuinely uncommon in this space.

## 5. Dataset Availability

Public datasets: identical to the free OHLCV sources already validated in prior FYP analysis for this team (yfinance, Alpha Vantage free tier), used here as calibration/validation targets rather than supervised-training labels. Data collection strategy: simple API-pull, no manual collection needed, this is a data-abundant, low-risk project on the data-access dimension specifically. Estimated dataset size needed: multi-year daily OHLCV history for a handful of assets/market periods (including at least one well-known historical stress event, e.g., a known flash-crash or crisis period, for the qualitative validation case study) is entirely sufficient.

## 6. Development Complexity

Difficulty: Medium (individually, the agent-rule design, the calibration/optimization loop, and the real-time visualization are each tractable; the main complexity is integration and, especially, getting the live-animation frontend smooth and performant, a genuine but well-precedented web-engineering challenge). Estimated completion time: 8 months. GPU requirements: essentially none; this entire project is CPU-bound (rule-based agent simulation and classical optimization, no deep-learning training loop at all), making it the single most compute-light idea on this entire list, a genuine advantage if the team is GPU-constrained or anxious about compute access. Team size suitability: 3 students (one on the agent-archetype simulation engine, one on the calibration/optimization pipeline and stylized-facts evaluation, one on the real-time interactive frontend, which is itself a substantial enough task to warrant a dedicated owner given the animation-quality bar this idea's demo value depends on).

## 7. Funding Potential

Why investors/incubators/grants may care: financial-literacy and investor-education tooling is a recognized, fundable category (both commercially, as an EdTech-adjacent fintech product, and via grant/foundation funding focused on financial literacy specifically), and the genuinely novel, highly-demoable nature of this product (few competitors offer anything visually similar) gives it strong pitch-competition and demo-day appeal beyond its grant/funding fit. Potential market size: large in the financial-literacy/EdTech-fintech crossover space, plus a plausible secondary market in finance-education institutions (business schools, online finance-course platforms) wanting an interactive teaching tool, a market this team's department itself is a natural first customer/showcase for. Commercialization opportunities: a freemium consumer education tool, or a B2B licensing angle selling the simulation engine to finance-education platforms/business schools as an interactive teaching aid, a credible and underexplored niche.

## 8. Research Potential

Good, with a genuinely distinctive angle. A paper on "calibrating agent-based market simulations to empirical stylized facts via simulation-based optimization: an accessible framework" fits the established agent-based computational finance research literature and could target a computational-finance or complex-systems-in-finance workshop (e.g., associated



with the Computational Social Science Society’s relevant tracks, or a finance-AI venue like ICAIF given its breadth). Research questions: what minimal agent-population-mix and behavioral-parameter configuration is sufficient to reproduce a target market period’s full stylized-facts profile (a genuinely interesting model-parsimony question); how sensitive are the simulation’s emergent flash-crash dynamics to small changes in the panic-seller archetype’s behavioral-threshold parameter, a clean, fully-controllable sensitivity-analysis question well-suited to this kind of simulation framework.

## 9. Risks and Challenges

Technical risks: calibration/optimization over a simulation can be slow if each simulation run is expensive to compute; the team should profile and optimize the simulation’s per-run compute cost early, since the calibration loop may need many hundreds of simulation runs to converge well. Data risks: minimal, this is a data-abundant project. Deployment risks: low; this is explicitly an educational/exploratory simulation tool, not a real-money trading or forecasting system, and this framing (no real predictions about actual future markets, purely an educational simulation of *how* markets can behave) should be stated clearly and prominently in the tool itself, both for honesty and to avoid any unintended “this predicts the market” misreading.

## 10. Final Verdict

Innovation: 8/10. Practicality: 8/10. Funding Potential: 6/10. Research Potential: 7/10. Ease of Completion: 8/10.

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## IDEA 8: Cross-Platform Coordinated Inauthentic Behavior Visualizer (Non-Deepfake Media Forensics / Agentic AI)

### 1. Problem Statement

Coordinated inauthentic behavior (bot networks, astroturfing campaigns, coordinated harassment or disinformation pushes) is a distinct and growing problem from individual deepfake content, and detecting it requires analyzing *network and timing patterns across many accounts* rather than analyzing any single piece of content in isolation, a fundamentally graph-and-temporal-pattern problem rather than a per-item classification problem, and most accessible tooling for this either doesn’t exist publicly or is locked inside platforms’ own internal trust-and-safety systems. This affects platform trust-and-safety teams, researchers studying disinformation, and journalists investigating coordinated campaigns, and matters because the detection signal here (network structure and timing synchrony across accounts) is structurally different from, and complements rather than duplicates, content-level deepfake detection.

## 2. Proposed Solution

A system that ingests a set of public social-media posts/accounts (using only publicly available data, respecting platform terms of service, e.g., via a research-oriented public dataset rather than live scraping, detailed below) and builds an interaction graph (who posts similar content, who retweets/amplifies whom, posting-time-synchrony patterns across accounts), then applies graph-based anomaly detection to flag clusters of accounts exhibiting statistically improbable coordination (near-identical posting times, near-identical text content, unusually tight mutual-amplification patterns), visualized as an explorable, interactive network graph where flagged clusters are highlighted and clickable for evidence drill-down.

The visual payoff is excellent and specifically suited to your interactivity requirement: a live, force-directed, explorable network graph (using a library like D3.js or a similar graph-visualization tool) where the user can zoom, pan, click on a flagged cluster to see the supporting evidence (a timeline of synchronized posts, a text-similarity heatmap among cluster members), a genuinely impressive, professional-looking “data investigation” interface for a viva demo.

## 3. Technical Architecture

Data sources: public, research-oriented social-media datasets specifically released for studying coordinated inauthentic behavior and bot networks (several academic disinformation-research datasets exist and are released specifically for this kind of research, typically anonymized/aggregated public posts with known ground-truth bot/coordinated-campaign labels from platforms’ own past disclosed takedowns, which is the standard, ethically-clean data source for this research area rather than live scraping).

ML/DL models: a graph-construction pipeline (building an account-interaction graph from posting/amplification/similarity relationships, a data-engineering task more than a modeling task); a graph neural network or simpler graph-statistics-based anomaly-detection model (e.g., a GNN-based node/cluster anomaly scorer, or, as a strong and more interpretable alternative given the project’s explainability needs, classical graph-clustering plus statistical-anomaly-scoring on cluster-level synchrony metrics, again following the lesson from the broader 2026 literature that simpler, interpretable methods often compete well with larger models and are easier to explain in a viva) for flagging coordinated clusters; a text-similarity model (a sentence-embedding model, e.g., a lightweight sentence-transformer, modest compute) for the content-similarity dimension of the coordination signal.

Training pipeline: if using a GNN-based approach, train on the labeled academic coordinated-behavior dataset with known ground-truth bot/campaign account labels; if using the classical graph-statistics approach (a reasonable, lower-risk, more interpretable alternative the team should seriously consider given the explainability and viva-demo value of a transparent method), the “training” is primarily threshold/parameter calibration against the labeled dataset rather than a deep training loop, an appropriately-scoped choice.

Evaluation pipeline: standard cluster-detection/anomaly-detection metrics (precision/recall of flagged accounts against the dataset’s known ground-truth coordinated-campaign labels); a qualitative case-study walkthrough (the team selects a known, labeled historical coordinated campaign from the dataset and shows, step by step, how their system’s graph-visualization

would have surfaced it, a genuinely strong, narrative-friendly evaluation approach for a viva demo).

Deployment architecture: a web app (React frontend with a graph-visualization library, FastAPI backend) where a user loads a dataset (a provided sample dataset for the demo, given that live data collection raises platform-ToS and privacy considerations the team should avoid for an FYP), explores the interactive network graph, and drills into flagged clusters' supporting evidence.

#### 4. Novelty Analysis

Existing solutions: this is an active area within computational social science and disinformation research (with established academic literature on bot-detection and coordinated-behavior-detection specifically), but accessible, open-source, visually-interactive tools for exploring this kind of analysis are not common outside specialized research labs or platforms' internal (non-public) tooling; most published academic work in this space presents results as static tables/graphs in a paper, not as an explorable interactive tool. Your contribution: building a genuinely interactive, explorable visualization layer on top of established coordinated-behavior-detection techniques, explicitly aimed at making this kind of investigative analysis accessible to a non-specialist (journalist, smaller platform's trust-and-safety team) audience rather than only to researchers with custom internal tooling, which is a legitimate and underexplored accessibility-focused contribution even where the underlying detection technique itself draws on established methods.

#### 5. Dataset Availability

Public datasets: the team must specifically search for and validate access to an academic disinformation/coordinated-behavior research dataset before committing to this idea (availability and access terms vary and some require a formal academic-research-use request, similar in friction to the deepfake-video datasets discussed in Ideas 1-2, a known and manageable process, not a blocker, but one requiring early action). Data collection strategy: the team should not attempt live social-media scraping for this project, both due to platform-ToS risk and ethical/privacy considerations around analyzing real, non-research-consented accounts; relying on an existing, ethically-sourced academic research dataset is the correct and only recommended approach. Estimated dataset size needed: academic coordinated-behavior datasets vary in scale, but a subset covering a few known historical campaigns with a few thousand to tens of thousands of involved accounts/posts is a realistic, sufficient target for this project's graph-analysis scale.

#### 6. Development Complexity

Difficulty: Medium-Hard (the graph-construction and interactive-visualization engineering is substantial, even though the underlying detection-modeling choice can be kept appropriately simple/interpretable per the guidance above). Estimated completion time: 8-9 months, contingent on early dataset-access validation. GPU requirements: low; if using a GNN, a single consumer GPU is more than sufficient given the modest graph scale involved; if using the classical graph-statistics approach (the recommended lower-risk path), GPU is barely needed at all beyond the lightweight sentence-embedding model for the text-similarity component. Team

size suitability: 3-4 students (one on the graph-construction/data-engineering pipeline, one on the anomaly-detection modeling choice, one on the interactive graph-visualization frontend, which is a substantial enough undertaking to warrant a dedicated owner, one floating on evaluation/dataset-validation).

## 7. Funding Potential

Why investors/incubators/grants may care: trust-and-safety and disinformation-research tooling is a recognized, internationally-fundable civic-tech and deep-tech category (similar funding landscape to the misinformation-tracker idea noted in prior FYP analysis for this team: international press-freedom/digital-rights funders, platform-integrity-focused grant programs), more grant/NGO-oriented than a fast-revenue VC pitch in the near term, a distinction worth stating honestly as with other media-integrity-focused ideas on this list. Potential market size: a specialized but real market, smaller platforms' trust-and-safety teams, disinformation researchers, journalism-integrity organizations, all genuine institutional buyers even if smaller in aggregate than a consumer-product market. Commercialization opportunities: licensed as an investigative tool to smaller platforms (large platforms have their own internal tooling and are unlikely customers, a realistic market-sizing caveat worth stating) and to journalism/research organizations, or positioned as an open-source research tool that builds the team's research/portfolio credibility even without direct near-term revenue.

## 8. Research Potential

Good. A paper on "accessible, interactive graph-based detection and visualization of coordinated inauthentic behavior" fits the active computational-social-science/disinformation-research literature, with realistic target venues including a workshop at ICWSM (the International Conference on Web and Social Media, a strong, undergraduate-reachable fit for exactly this kind of work) or a security/trust-and-safety-focused workshop. Research questions: which graph-structural and timing-synchrony features are most predictive of confirmed coordinated campaigns versus organic viral content (a genuinely important, currently-debated distinction in this research area, since organic virality can superficially resemble coordination); how well does a detection method calibrated on one historical campaign's coordination pattern generalize to a structurally different, later campaign, a generalization question structurally similar in spirit to the cross-dataset rigor emphasized across Ideas 1-3 on this list.

## 9. Risks and Challenges

Technical risks: distinguishing genuine coordinated inauthentic behavior from organic viral/trending content is a known, genuinely hard problem in this research area, and the team should expect and honestly report false-positive cases (organic trends that superficially look coordinated) rather than presenting the system as if this distinction were fully solved. Data risks: dataset access is the primary friction point and must be validated early; live scraping should be explicitly avoided for the ethical and ToS reasons stated above. Deployment risks: a tool that incorrectly flags genuine organic activity as "coordinated inauthentic behavior" could itself contribute to false accusations against real accounts/communities; the tool must be framed and demoed explicitly as an investigative-assistance/hypothesis-generation aid for a human analyst to verify, never as an automated accusation system, and this framing should be visible in the UI itself.

## 10. Final Verdict

Innovation: 7/10. Practicality: 6/10. Funding Potential: 5/10. Research Potential: 7/10. Ease of Completion: 5/10.

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## IDEA 9: Explainable Anomaly Detection for DeFi (Decentralized Finance) Smart Contract Transaction Patterns (Non-Banking FinTech)

### 1. Problem Statement

DeFi (decentralized finance) protocol users and smaller DeFi platform operators have very limited tooling to detect anomalous, potentially exploitative transaction patterns (flash-loan attacks, unusual liquidity-pool drain patterns, wash-trading-style volume manipulation) in real time, because most rigorous on-chain security analysis is either manual (expensive specialist audits, done periodically, not continuously) or locked inside large, well-funded protocols' internal tooling, leaving smaller DeFi projects and individual users without accessible, continuous anomaly monitoring. This affects smaller DeFi protocol teams, individual DeFi users managing their own positions, and on-chain security researchers, and matters because DeFi exploits are a well-documented, costly, ongoing problem and continuous, accessible anomaly monitoring (versus periodic manual audit) is a genuine, currently underserved capability gap, especially for smaller, less-resourced protocols.

### 2. Proposed Solution

A system that ingests public on-chain transaction data for a given DeFi protocol/smart contract (via free blockchain-explorer or node-provider APIs, entirely public data, no special access needed since blockchain data is inherently public by design) and applies time-series and graph-based anomaly detection to flag statistically unusual transaction patterns in near-real-time, presenting findings via an interactive dashboard showing the specific transaction graph, the anomaly score over time, and a plain-language explanation of which specific feature(s) (transaction-size spike, unusual sender-pattern, liquidity-ratio shift) triggered the flag.

This is visually compelling and demoable: a live transaction-flow graph (showing token movement between addresses as an animated, flowing visualization) with an anomaly-score timeline beneath it, and when an anomaly is flagged, the specific suspicious transaction(s) highlighted directly on the flow graph with the triggering feature labeled, giving a genuinely "watch the AI catch something suspicious happen" demo moment.

### 3. Technical Architecture

Data sources: public blockchain data via free node-provider/explorer APIs (e.g., free-tier access to a major blockchain's transaction data, entirely public and requiring no special permissions since blockchain ledgers are public by design, a genuine data-access advantage over almost every other idea on this list) for a chosen DeFi protocol or set of protocols (the team should pick a few well-known, sufficiently active DeFi protocols with public, well-documented smart-

contract interfaces to focus on, rather than attempting protocol-agnostic generality from day one).

ML/DL models: an unsupervised anomaly-detection model over engineered transaction-feature time series (transaction-size distribution shifts, liquidity-pool-ratio changes, sender/receiver-pattern novelty, all computed from the raw on-chain data), using either a statistical approach (e.g., an Isolation Forest or a simpler rolling-statistics-based anomaly score, both lightweight, interpretable, and well-suited to the explainability requirement) or, as a more sophisticated option if the team wants additional technical depth, a graph-based anomaly-detection model treating the transaction network itself as a graph and flagging structurally anomalous sub-patterns (e.g., a flash-loan attack's characteristic single-transaction-block borrow-manipulate-repay pattern, which has a genuinely distinctive graph signature worth specifically targeting as a named attack pattern, not just generic anomaly detection).

Training pipeline: since real labeled DeFi-exploit data is sparse (most known exploits are well-documented individually but don't form a large labeled training set), the team should combine the limited number of well-documented, publicly post-mortemed historical DeFi exploits (several major exploits have detailed public post-mortem write-ups describing the exact transaction pattern, providing genuine, if few, real labeled positive examples) with predominantly unsupervised anomaly-detection methods (which don't require large labeled-attack training sets, an appropriate and honest methodological choice given the real data-scarcity constraint here) calibrated on normal-operation transaction data as the primary training signal.

Evaluation pipeline: the team should specifically test their system against the handful of well-documented historical DeFi exploits with public post-mortems (a clean validation approach: "would our anomaly detector have flagged this known, real, costly exploit before/as it happened," a genuinely compelling and honest evaluation framing) alongside standard unsupervised-anomaly-detection evaluation on normal-versus-synthetically-injected-anomalous transaction patterns the team constructs themselves (e.g., simulating a flash-loan-attack-pattern transaction sequence and checking detection).

Deployment architecture: a web dashboard (React/FastAPI) with the live transaction-flow graph visualization, anomaly-score timeline, and flagged-transaction drill-down described above; given that true real-time on-chain monitoring requires sustained API usage, the demo can reasonably use either a live-but-rate-limited real-time feed or a "replay" mode stepping through historical data (including the documented historical exploits) at a controllable pace for the viva demo, both legitimate and clearly-labeled deployment modes.

## 4. Novelty Analysis

Existing solutions: on-chain security/auditing firms exist and do sophisticated analysis, but their tooling is largely proprietary, expensive, and aimed at well-funded protocols as clients, not accessible to smaller projects or individual users; some open-source on-chain-analytics tools exist for general blockchain exploration but are rarely purpose-built for continuous, explainable anomaly flagging with an accessible visual interface. Your contribution: an accessible, explainable, continuously-monitoring anomaly-detection tool aimed specifically at the underserved smaller-protocol/individual-user segment, validated explicitly against real,



documented historical exploits (not just synthetic anomalies), a genuine and creditable validation approach most accessible/open tools in this space don't bother to do rigorously.

## 5. Dataset Availability

Public datasets: blockchain transaction data itself is entirely public and freely accessible via standard node-provider/explorer APIs, the single best data-access situation on this entire list, no application/approval process needed at all, unlike the deepfake-video and coordinated-behavior datasets discussed elsewhere in this report. Data collection strategy: direct API-pull of historical and live transaction data for the chosen protocol(s); the historical-exploit validation cases require the team to research and reconstruct the specific transaction sequences from public post-mortem write-ups and on-chain explorers, a legitimate research-and-data-engineering task. Estimated dataset size needed: months of historical transaction data for a few chosen DeFi protocols (easily available, often hundreds of thousands of transactions, far more than needed) plus the handful of specifically-reconstructed historical-exploit case studies; this is a data-abundant project on the normal-operation side and appropriately data-scarce (and honestly treated as such) on the labeled-exploit side.

## 6. Development Complexity

Difficulty: Medium. Estimated completion time: 8 months. GPU requirements: very low; this entire project is comfortably CPU-feasible (Isolation Forest, rolling statistics, and even a modestly-sized graph-anomaly model all run without GPU acceleration at this data scale), another genuine compute-light option for a GPU-constrained team. Team size suitability: 3 students (one on the on-chain data-engineering pipeline, one on the anomaly-detection modeling and historical-exploit validation, one on the live dashboard/visualization).

## 7. Funding Potential

Why investors/incubators/grants may care: DeFi/blockchain-security tooling is a well-established, internationally fundable category (on-chain security is a recognized, well-funded niche within the broader crypto/Web3 investment landscape), and accessible/affordable security monitoring for smaller protocols is a credible, underserved market wedge against expensive incumbent audit firms. Potential market size: global and growing alongside the broader DeFi ecosystem; smaller DeFi protocols, individual large DeFi position-holders, and on-chain security researchers are all plausible customer segments. Commercialization opportunities: a subscription-based continuous-monitoring SaaS for smaller DeFi protocols (the most realistic, currently-fundable angle, directly analogous to established Web3-security-monitoring startups, giving the team credible market comparables to reference), or an open-source core tool with a paid premium-monitoring tier.

## 8. Research Potential

Moderate-to-good. A paper on “explainable, unsupervised anomaly detection for DeFi transaction patterns validated against historical exploit case studies” is a reasonable contribution at the intersection of applied ML and blockchain/Web3 security research, potentially fitting a blockchain-security-focused workshop or a broader applied-anomaly-detection venue. Research questions: how early, relative to the actual exploit transaction, would

the system's anomaly score have crossed an alerting threshold for each documented historical exploit (a genuinely valuable, specific, and honestly-answerable “would this have helped in practice” question); does a graph-structural approach meaningfully outperform a simpler feature-based statistical approach specifically for flash-loan-attack-pattern detection, a clean, fundable architectural comparison.

## 9. Risks and Challenges

Technical risks: unsupervised anomaly detection is prone to false positives on legitimate but unusual activity (e.g., a large legitimate trade during high volatility might resemble an attack pattern statistically); the team must report false-positive rates honestly, not just true-positive catches on the known historical exploits. Data risks: minimal, blockchain data access is the most friction-free data source on this entire list. Deployment risks: a missed real exploit (false negative) or a wrongly-flagged legitimate large transaction (false positive, which could itself spook users/cause reputational harm to a legitimate large trader) both carry real consequences; the tool should be framed as a monitoring/alerting aid for a human security team to investigate further, not an autonomous halt-the-protocol system, which is both the honest framing and the safer one.

## 10. Final Verdict

Innovation: 7/10. Practicality: 7/10. Funding Potential: 7/10. Research Potential: 6/10. Ease of Completion: 7/10.

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# IDEA 10: Lightweight Cross-Modal (Audio-Visual Sync) Deepfake Detector Targeting Lip-Sync Inconsistency at Scale

## 1. Problem Statement

A large and growing share of real-world deepfakes are talking-head/lip-sync manipulations (a face is kept largely intact but the mouth movement is re-synced to different audio, including AI-dubbed or voice-cloned audio), and per the 2025 Celeb-DF++ benchmark, this “talking-face” category is both growing (now representing about half of that benchmark’s generated forgeries) and specifically harder to detect than classic face-swap, with detection rates degrading substantially on this category compared to face-swap. This affects the same broad population as Idea 1 (platforms, investigators, fraud-prevention teams), with a specific emphasis on the financial-fraud angle (voice-cloned, lip-synced executive-impersonation videos are a documented and growing corporate-fraud vector), and matters because this specific, currently-hardest-to-detect category deserves dedicated, focused attention rather than being treated as a minor sub-case of general deepfake detection.

## 2. Proposed Solution

A lightweight cross-modal detector specifically targeting audio-visual synchrony at the phoneme/viseme level (does the mouth shape match what the audio claims is being said, at a

fine temporal grain), rather than general-purpose face-swap artifact detection, explicitly because the Celeb-DF++ findings show this talking-face category is the field's current weak point. The system extracts lip-region visual features and audio phonetic features in parallel, and scores their temporal alignment, flagging videos where the mouth-shape sequence is statistically implausible given the spoken audio.

The visual payoff is strong and intuitive: a side-by-side display of the extracted mouth-region video crop and a phoneme-timeline extracted from the audio, with a live synchrony-score curve showing exactly where in the video the lip-audio alignment breaks down, an interpretable, fine-grained, moment-by-moment visualization that's both technically substantive and immediately understandable to a non-expert viva audience ("see here, right at this timestamp, the mouth doesn't match the sound").

### 3. Technical Architecture

Data sources: FaceForensics++ and Celeb-DF++ specifically filtered/subset to their talking-face and lip-sync-relevant manipulation categories (Celeb-DF++ explicitly provides this talking-face category as one of its three forgery scenarios, directly usable for this focused sub-problem), supplemented by any available dedicated lip-sync-deepfake datasets (e.g., datasets built around well-known lip-sync-generation methods, several of which have research datasets released alongside their papers).

ML/DL models: a lip-region visual feature extractor (a lightweight CNN or a pretrained lip-reading model's visual front-end, several open lip-reading models exist and are modest in compute requirement for feature extraction/fine-tuning) paired with an audio phonetic-feature extractor (a pretrained speech model's phoneme/viseme-relevant intermediate features, e.g., from a wav2vec2-style model, reused similarly to Idea 3's audio feature-extraction approach); a lightweight cross-modal alignment-scoring model (a small Transformer or even a simpler cross-correlation-based alignment-scoring approach between the two feature streams, again following the explicit 2026-literature lesson favoring small, targeted models over large backbones) trained to distinguish genuinely-synced from artificially-resynced audio-visual pairs.

Training pipeline: extract paired lip-region-visual and audio-phonetic feature sequences from the training videos (a meaningful preprocessing investment, similar in spirit to Idea 1's rPPG-extraction preprocessing step, and worth budgeting real time for); train the alignment-scoring model to output high synchrony scores for genuine pairs and low scores for resynced/manipulated pairs, using the labeled real-versus-talking-face-manipulated videos from the training data; explicit, deliberate focus on this specific manipulation category rather than general-purpose multi-category training, since the project's entire value proposition is depth on this one currently-underperforming category, not breadth.

Evaluation pipeline: detection AUC specifically on Celeb-DF++'s talking-face category test split (directly, honestly comparable to that benchmark's own published baseline numbers for the same category, since the 2025 paper itself reports degraded detection on this specific category, giving your team's result a clear, citable point of comparison); cross-dataset generalization test against any additional available lip-sync-specific dataset; an explicit ablation comparing cross-modal audio-visual alignment scoring against a visual-only baseline (do you actually need the

audio-modality information, or does visual-only lip-region analysis already catch most cases, a clean, directly informative ablation either way the result lands).

Deployment architecture: a web app (React/FastAPI) with the side-by-side mouth-crop-video and phoneme-timeline display, the live synchrony-score curve, and a final verdict with the specific flagged-timestamp evidence highlighted, directly fulfilling both the visual-interactivity and explainability requirements.

#### 4. Novelty Analysis

Existing solutions: general-purpose deepfake detectors (including most face-swap-era detectors and even broad rPPG-based approaches like Idea 1) are not specifically optimized for the talking-face/lip-sync category, and the 2025 Celeb-DF++ paper explicitly documents that while face-swap forgeries are detected robustly (AUC above 85%), face-reenactment and talking-face forgeries yield substantially degraded detection rates in the AUC 50-70% range, a clear, current, citable weak point in the field; some lip-sync-detection research exists specifically (it's not an entirely unexplored sub-problem) but is less mature and less benchmarked than face-swap detection broadly. Your contribution: a deliberately narrow, deep focus on exactly this documented current weak point, with rigorous benchmarking specifically against the category where the field's own most recent large benchmark shows the most room for improvement, a sharply-targeted, honestly-scoped, and currently well-justified research focus.

#### 5. Dataset Availability

Public datasets: FaceForensics++ and Celeb-DF++ (the same access process as Ideas 1-2, academic-use request, standard and fast), with Celeb-DF++ specifically providing the talking-face category subset needed here. Data collection strategy: no additional manual collection strictly needed, though the team could supplement with their own simple lip-sync test cases (using open lip-sync-generation tools to create additional test examples) for extra robustness testing at low cost. Estimated dataset size needed: the talking-face subset of Celeb-DF++ alone (a meaningful fraction of its reported ~53,000 fake clips across 22 methods) provides ample data for this focused sub-problem; the team does not need the dataset's full breadth, only its talking-face-relevant portion.

#### 6. Development Complexity

Difficulty: Medium (genuinely more tractable than Ideas 1, 2, or 3 on this list because the scope is deliberately narrower and the feature-extraction backbones, lip-reading and speech-phonetic models, are mature, pretrained, and well-documented, reducing from-scratch modeling risk). Estimated completion time: 7-8 months, making this one of the more comfortably-timed ideas on the list. GPU requirements: low-to-moderate; feature extraction using pretrained lip-reading and speech models is comfortable on a single consumer GPU (8GB+) or free Colab/Kaggle GPU, and the cross-modal alignment-scoring model itself is small, consistent with the broader lightweight-model lesson from the current literature. Team size suitability: 3 students (one on visual lip-region feature extraction, one on audio phonetic feature extraction and the cross-modal alignment model, one on the dashboard/visualization).

## 7. Funding Potential

Why investors/incubators/grants may care: this targets a documented, currently-underperforming, and financially-relevant deepfake category (executive-impersonation voice-and-video fraud specifically often relies on exactly this talking-face/lip-sync manipulation type), giving it a sharp, currently-relevant security/fraud-prevention pitch; fits the same deep-tech grant landscape (Ignite, HEC Technology Development Fund) as Ideas 1-3. Potential market size: global, and the narrow technical focus makes for an easier, more specific B2B pitch (“we specifically catch the lip-sync category general detectors miss”) than a broad “we detect all deepfakes” claim would be, since specificity is often more credible to a technical buyer than an overly broad claim. Commercialization opportunities: positioned as a complementary module to a broader deepfake-detection suite (a natural pairing with Idea 1’s rPPG approach, since the two target different, complementary manipulation categories, face-swap/reenactment versus talking-face/lip-sync, a credible “stack these two together” product narrative if the department has multiple teams pursuing related ideas).

## 8. Research Potential

Strong, and arguably the most immediately citable idea on this list given how directly it engages with a just-published (2025) benchmark’s explicitly-stated weak point. A paper on “closing the talking-face detection gap: cross-modal audio-visual synchrony scoring for lip-sync deepfake detection” is directly publishable, with a clean, current point of comparison already established by Celeb-DF++’s own reported baseline numbers; realistic target venues include WIFS, an ICASSP audio-visual-forensics track, or a CVPR/WACV media-forensics workshop. Research questions: how much of the talking-face detection gap closes with cross-modal (audio+visual) alignment scoring specifically, versus what a strong visual-only lip-region model alone already achieves (the project’s central, directly fundable ablation); does performance hold up specifically against voice-cloned (not just re-dubbed/re-timed) audio paired with lip-sync video, a meaningfully harder and more current sub-case worth testing explicitly if time permits.

## 9. Risks and Challenges

Technical risks: cross-modal alignment scoring requires careful temporal synchronization of the two feature streams during preprocessing, a genuine, fiddly engineering detail that’s easy to get subtly wrong; the team should validate this preprocessing step carefully with sanity checks (e.g., on clearly genuine videos, synchrony scores should be uniformly high) before trusting downstream results. Data risks: minimal beyond the standard academic-dataset-access process already discussed for Ideas 1-2. Deployment risks: low, same forensic-assistance, human-in-the-loop framing as Idea 1 applies here, and should be stated the same way in both the UI and final report.

## 10. Final Verdict

Innovation: 8/10. Practicality: 8/10. Funding Potential: 7/10. Research Potential: 8/10. Ease of Completion: 8/10.

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## Cross-Idea Ranking and Comparison

### Overall Ranking (Best to Worst for an Undergraduate BSAI FYP)

- 1. Idea 10 — Lip-Sync/Talking-Face Cross-Modal Detector.** The best overall balance on this entire list: a deliberately narrow, currently well-justified research focus (directly engaging a documented 2025 benchmark weak point), mature pretrained building blocks reducing modeling risk, a comfortable 7-8 month timeline, strong visual demo value, and a sharp, specific, credible commercial pitch. This is the safest high-reward choice on the list.
- 2. Idea 1 — Multi-Region Physiological Synchrony Detector.** Your original inspiration, strengthened with explicit 2026 research grounding and rigorous cross-dataset evaluation; excellent visual demo value (live animated waveforms), strong research story, slightly more technically ambitious than Idea 10 but well within reach for a solid team.
- 3. Idea 7 — Agent-Based Market Regime Stress-Test Sandbox.** The most visually spectacular demo on the entire list (live animated multi-agent market simulation), zero GPU requirement, comfortable timeline, and a genuinely novel accessibility angle on an established academic technique; slightly weaker pure-publication story than the top two deepfake ideas, but exceptional practicality and demo impact.
- 4. Idea 3 — Modern TTS Audio Deepfake Generalization Detector.** Extremely fresh, currently-open research gap (the January 2026 emotional-speech finding is genuinely new), strong funding relevance (voice-fraud is a hot, real threat), comfortable compute profile; slightly more data-engineering-dependent (team must reliably generate quality modern-TTS samples) than the top three.
- 5. Idea 9 — Explainable DeFi Anomaly Detection.** The most data-access-friendly idea on the entire list (blockchain data is unconditionally public), solid funding story in the Web3-security space, good visual demo (live transaction-flow graph); slightly less groundbreaking on pure research novelty than the top deepfake ideas.
- 6. Idea 4 — Multi-Agent Crypto Market Microstructure Visualizer.** Excellent practicality and demo value, comfortable for a team wanting strong software-engineering execution over deep research risk; weaker pure-publication angle than most other ideas, more of a strong-product, lighter-research pick.
- 7. Idea 5 — Agentic Personal Expense-Intelligence Simulator.** Genuinely novel and well-differentiated from generic budgeting apps, comfortable compute and timeline; weaker funding/market story than the institutional-buyer-focused FinTech ideas, since consumer-fintech monetization is harder, and a moderate (not top-tier) research angle.
- 8. Idea 6 — Synthetic Media Provenance Chain Visualizer.** Strong, genuinely novel investigative-tool concept; the provenance-ordering component carries real research risk and should be scoped carefully; recommended for a team comfortable with some result uncertainty in exchange for high novelty.
- 9. Idea 2 — Diffusion-Era Single-Frame Forensic Localizer.** The most frontier-research-adjacent, highest-novelty idea on the entire list, but also the hardest and riskiest; recommended



only for a strong team explicitly prioritizing research novelty over completion certainty, with a clear fallback plan from month 6.

**10. Idea 8 — Coordinated Inauthentic Behavior Visualizer.** Excellent concept and strong research/funding fit for the right team, but the most dataset-access-fragile idea on the list (academic coordinated-behavior datasets are less standardized in availability than the deepfake-video datasets) and the most engineering-heavy on the visualization side; recommended only if dataset access is validated very early.

### Top 3 Recommended Ideas (Best Balance of Innovation, Feasibility, Visual/Interactive Quality, Funding, and Research Potential)

Based on your explicit five-way balance requirement, the three strongest picks are **Idea 10 (Lip-Sync Cross-Modal Detector)**, **Idea 1 (Multi-Region Physiological Synchrony Detector)**, and **Idea 7 (Agent-Based Market Regime Stress-Test Sandbox)**. Detailed roadmaps follow.

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## TOP PICK #1 — Idea 10: Lip-Sync/Talking-Face Cross-Modal Deepfake Detector

### 12-Month Roadmap (Month-Wise)

**Month 1:** Request academic access to FaceForensics++ and Celeb-DF++ (start immediately, approval typically takes days to a couple of weeks); literature review specifically on lip-sync/audio-visual-deepfake-detection prior work and on the pretrained lip-reading and speech-phonetic models the team will reuse; set up the project repository and experiment-tracking.

**Month 2:** Build the data-filtering pipeline isolating Celeb-DF++'s talking-face category and FF++'s relevant manipulation types; build the lip-region face-tracking and cropping pipeline (MediaPipe or similar); validate this preprocessing on a small sample before scaling.

**Month 3:** Integrate the pretrained lip-reading visual-feature extractor and the pretrained speech-phonetic feature extractor (wav2vec2-style); extract and cache features for the full filtered training set, the single most time-consuming preprocessing step, budget real time here.

**Month 4:** Build and train the cross-modal alignment-scoring model on the extracted feature pairs; get a first working, even if mediocre, end-to-end pipeline before optimizing.

**Month 5:** Tune and improve the alignment model; run the visual-only ablation baseline (the lip-region model alone, without audio) for the planned comparison study.

**Month 6 (Checkpoint):** Full evaluation on the Celeb-DF++ talking-face held-out test split with direct comparison against the benchmark's own reported baseline numbers for this category; if results are weak, this is the point to revisit feature-extraction quality or model architecture, not push forward blindly.

**Month 7:** Build the cross-dataset generalization test (if a second lip-sync-relevant dataset is available) or, if not, design and run the team's own additional lip-sync test cases using open lip-sync-generation tools for supplementary robustness testing.

**Month 8:** Build the dashboard frontend (mouth-crop video display, phoneme-timeline visualization, live synchrony-score curve, flagged-timestamp evidence highlighting); integrate the trained model via the FastAPI backend.

**Month 9:** Polish the dashboard interactivity and visual presentation; run the full evaluation suite end-to-end and finalize all reported numbers, including the visual-only-versus-cross-modal ablation.

**Month 10:** Write the methodology and results sections rigorously; draft the research paper alongside the FYP report.

**Month 11:** Prepare the FYP defense demo and presentation materials; seek any available informal feedback from faculty before the final defense; if pursuing funding, begin the HEC Technology Development Fund or Ignite application process now.

**Month 12:** Final polish, defense, and submission of both the FYP report and, ideally, the drafted research paper to a target venue.

## Tech Stack

Python, PyTorch, a pretrained lip-reading model (visual front-end), wav2vec2 or similar pretrained speech model (HuggingFace Transformers), a lightweight Transformer or cross-correlation-based alignment-scoring model built in PyTorch, MediaPipe (face/lip tracking), FastAPI, React with a video/audio-visualization library (e.g., Wavesurfer.js for the audio-timeline component), deployed on a free-tier cloud host or university server.

## MVP Scope

A working pipeline that takes a video clip, extracts lip-region visual features and audio phonetic features, computes a frame-by-frame synchrony score, and flags the clip as genuine or lip-sync-manipulated with the specific lowest-synchrony timestamps highlighted; the cross-dataset generalization test and supplementary voice-cloned-audio testing are explicitly stretch goals, not core MVP requirements.

## Final Deliverables

The trained cross-modal alignment model with full experiment logs; the benchmark comparison against Celeb-DF++'s reported talking-face baseline numbers; the visual-only-versus-cross-modal ablation study; the deployed interactive dashboard; the FYP written report; a drafted research paper.

## Possible Research Paper Title

"Closing the Talking-Face Detection Gap: Cross-Modal Audio-Visual Synchrony Scoring for Lip-Sync Deepfake Detection"

## Startup Pitch

“[Product name] catches the lip-sync and voice-cloned deepfakes that general-purpose detectors consistently miss, the exact category executive-impersonation fraud relies on, with a live, explainable synchrony score pinpointing the precise moment a video’s mouth stops matching its audio.”

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## TOP PICK #2 — Idea 1: Multi-Region Physiological Synchrony Deepfake Detector

### 12-Month Roadmap (Month-Wise)

**Month 1:** Request academic access to FaceForensics++ and Celeb-DF/Celeb-DF++; download DeepFake-Eval-2024 (no application needed); literature review of rPPG-based deepfake detection (AVENUE, STFT-phase, Forensic Lens) to properly position the project’s contribution.

**Month 2:** Build the face-tracking and multi-region ROI extraction pipeline (forehead, left cheek, right cheek, optionally chin); validate ROI tracking stability on sample videos before scaling.

**Month 3:** Implement the CHROM/POS classical rPPG signal-extraction algorithm per region; extract and cache rPPG waveforms for the full training set, the slow, CPU-bound preprocessing step requiring real time budgeting.

**Month 4:** Build and train the lightweight temporal classifier (small TCN or compact Transformer) on the multi-region waveform and synchrony features; get a first working pipeline on FF++ before optimizing.

**Month 5:** Run the single-region-versus-multi-region-synchrony ablation (the project’s central empirical claim); tune the model based on these results.

**Month 6 (Checkpoint):** Cross-dataset evaluation against Celeb-DF++ and DeepFake-Eval-2024; this is the most important number in the entire project and should be run and honestly reported even if the result is humbling.

**Month 7:** Investigate performance breakdown by manipulation category specifically (face-swap versus reenactment versus talking-face, per Celeb-DF++’s categories), directly engaging the research question about category-specific generalization gaps.

**Month 8:** Build the live interactive dashboard (animated multi-region waveform display, synchrony-mismatch gauge, plain-language verdict explanation).

**Month 9:** Robustness testing under video compression (simulating real-world WhatsApp/social-media re-encoding); polish the dashboard.

**Month 10:** Write the methodology and results rigorously, with full honesty about the generalization-gap findings; draft the research paper.

**Month 11:** Prepare defense materials; begin any funding-application process if pursuing one.

**Month 12:** Final polish, defense, submission.

### Tech Stack

Python, OpenCV/MediaPipe (face tracking), classical rPPG algorithms (CHROM/POS, implementable directly or via existing open rPPG libraries), PyTorch (temporal classifier), FastAPI, React with a live-charting library (e.g., Chart.js or D3.js for the animated waveform display), deployed on a free-tier cloud host or university server.

### MVP Scope

A working pipeline that extracts multi-region rPPG waveforms from an uploaded video, computes synchrony-mismatch features, and outputs a real/fake classification with the synchrony score and per-region waveform visualization; full cross-dataset robustness-under-compression testing is the stretch-goal depth layer, not core MVP.

### Final Deliverables

The trained temporal classifier with full experiment logs; the single-region-versus-multi-region ablation results; the cross-dataset (FF++ to Celeb-DF++ to DeepFake-Eval-2024) generalization results; the deployed live interactive dashboard; the FYP written report; a drafted research paper.

### Possible Research Paper Title

“Multi-Region Physiological Synchrony as a Generalizable Deepfake Forensic Cue: A Cross-Dataset Evaluation”

### Startup Pitch

“[Product name] verifies video authenticity by checking what no deepfake generator currently models: whether your heartbeat shows up the same way, at the same time, across every part of your face.”

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## TOP PICK #3 — Idea 7: Agent-Based Market Regime Stress-Test Sandbox

### 12-Month Roadmap (Month-Wise)

**Month 1:** Set up the data pipeline (yfinance, Alpha Vantage) and cache historical OHLCV data for several assets and known historical stress-event periods; literature review of agent-based computational-finance modeling.

**Month 2:** Design and implement the agent-archetype rule-set (momentum-chaser, contrarian, panic-seller, fundamental-value-investor), keeping each archetype's decision rule simple and interpretable by design.

**Month 3:** Build the core simulation engine (agent population interacting to produce emergent price dynamics); validate basic sanity behavior (does a 100%-momentum-chaser population produce a trending/bubble-prone pattern, as theoretically expected) before adding complexity.

**Month 4:** Implement the stylized-facts evaluation framework (volatility clustering, fat tails, autocorrelation structure) for scoring how closely simulation output matches real historical market statistics.

**Month 5:** Build the calibration/optimization loop (via Optuna or a genetic algorithm) tuning agent-population-mix and behavioral parameters to minimize the gap between simulated and real stylized facts for a chosen target market period.

**Month 6 (Checkpoint):** Run the calibration to convergence on at least one real historical period; evaluate how well the calibrated simulation reproduces that period's actual statistical properties; this is the project's core empirical validation milestone.

**Month 7:** Run the flash-crash qualitative case study (shock the calibrated simulation with a sudden negative event and check whether the emergent crash-and-recovery pattern resembles a real historical flash-crash episode).

**Month 8:** Begin building the real-time interactive frontend (this is a substantial undertaking and deserves its own dedicated month-plus of focus given the demo-value the animation quality drives); prioritize getting smooth, responsive live animation working over adding more agent-archetype complexity.

**Month 9:** Integrate the population-mix sliders and live re-simulation triggering; integrate the side-by-side stylized-facts comparison panel.

**Month 10:** Full polish pass on the interactive frontend and simulation responsiveness; run the complete evaluation suite and finalize all reported results.

**Month 11:** Write the methodology and results; draft the research paper; prepare defense materials and, if pursuing the EdTech/financial-literacy commercialization angle, a separate pitch-deck summary for that audience specifically.

**Month 12:** Final polish, defense, submission.

## Tech Stack

Python (simulation engine, calibration logic), Optuna (calibration/hyperparameter optimization), pandas/numpy, yfinance/Alpha Vantage (data sourcing), FastAPI, React with a canvas/WebGL-capable visualization layer for smooth live animation (or a Plotly/Dash-based alternative if the team prioritizes faster development over animation polish), deployed on a free-tier cloud host or university server.

## MVP Scope

A working simulation engine with the four agent archetypes, a functioning calibration loop reproducing at least one real historical period's basic stylized facts, and an interactive frontend with population-mix sliders and live (even if not perfectly smooth) re-simulation and chart redraw; full WebGL-quality animation polish and the flash-crash case study are valuable depth additions but not strictly required for a defensible MVP.

## Final Deliverables

The calibrated simulation engine with full experiment logs; the stylized-facts validation results against real historical market data; the flash-crash qualitative case study; the deployed interactive dashboard with live agent-population-mix exploration; the FYP written report; a drafted research paper.

## Possible Research Paper Title

"Calibrating Agent-Based Market Simulations to Empirical Stylized Facts via Simulation-Based Optimization: An Accessible Framework"

## Startup Pitch

"[Product name] lets anyone, not just institutional quants, watch a market crash and recover in real time by adjusting trader-behavior sliders, turning abstract market-microstructure theory into something you can see and play with."

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## Funding-Channel Map: Which Ideas Attract Which Funding Source

**FinTech innovation funding (non-banking specifically, as required):** Ideas 4, 5, 7, and 9 are your non-banking FinTech options, and all four cleanly avoid any banking/credit-scoring entanglement; Idea 9 (DeFi anomaly detection) has the sharpest, most currently-fundable Web3-security angle internationally, while Idea 7 (market simulation sandbox) has the strongest financial-literacy/EdTech-crossover funding story and the best demo-day pitch-competition appeal of the group.

**Government and deep-tech grants (Ignite, HEC Technology Development Fund):** Ideas 1, 2, 3, and 10 (the core deepfake/media-forensics cluster) are the strongest fits given their genuine algorithmic novelty and clear university-research-to-commercialization narrative; Idea 9's Web3-security angle is also a credible deep-tech grant fit.

**AI incubators / startup pathways (Ignite NICs, Plan9/PlanX):** Idea 10 (sharpest, most specific commercial pitch), Idea 1 (strong general deepfake-detection market), Idea 7 (best demo-day/pitch-competition material), and Idea 9 (clear Web3-security SaaS path) are the strongest incubator-pitch candidates; Idea 4 is also a solid, lower-risk incubator candidate given its strong practicality score.



**University research / publication pathways:** Ideas 1, 2, 3, 6, 8, and 10 (everything in the media-forensics/deepfake cluster, plus the coordinated-inauthentic-behavior idea) have the strongest pure-publication stories, since this entire space is genuinely active, current research territory with real open gaps your team's work could plausibly contribute to, not a saturated area where only incremental replication is possible.

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## Closing Recommendation

If forced to choose one idea optimizing simultaneously for FYP-evaluator approval, completion certainty, visual/interactive demo quality, funding access, and publication credibility, **Idea 10 (Lip-Sync Cross-Modal Detector)** is the single best-balanced choice on this list: it is narrowly and currently well-justified (directly engaging a documented 2025 benchmark weak point you can cite by name), built substantially from mature pretrained components that reduce from-scratch modeling risk, comfortably timed at 7-8 months, and produces an immediately legible, technically substantive demo. **Idea 1** is the closest alternative and the most faithful, strengthened evolution of your own original inspiration, while **Idea 7** stands out specifically if your team wants the single most visually spectacular, audience-impressing demo on this entire list, with the tradeoff of a somewhat less conventional publication story.

A final honest note, consistent with the same principle from any rigorous FYP evaluation: every deepfake-detection idea on this list will, if executed rigorously, surface a real, possibly humbling generalization gap when tested cross-dataset, that is in fact the entire point and the most current, citable finding in this field right now. The team that reports this gap honestly, rather than only showcasing the easy in-dataset number, will be more credible to both an academic committee and a deep-tech funder than a team that only shows a single inflated accuracy figure.